Tang Kinship under the Test of the 64→20 Invariant : Algebraic Topology, Alliance Filtration, and Social Homeostasis

Bruno DE DOMINICIS

June 2026

Abstract : This article demonstrates that the marriage rules and alliance structure under the Tang dynasty (notably the "Five Names and Seven Clans" system) are not cultural arbitraries, but obey the universal topological constraint of the 64→20 invariant. By modeling the space of relational possibilities via Clifford algebra Cl(6,0) and its 64 configurations (isomorphic to the 64 hexagrams of the *Yijing*), we show that nilpotence and the geometry of the Merkabah filter these 64 states to retain only 20 stable equivalence classes (the alliance attractors). This topological filtration, which prohibits "frustrated" configurations (incest taboos), ensures the negentropic homeostasis of the Chinese social system, in radical contrast to the Western schism analyzed by Viguier. **DOI :** 10.5281/zenodo.xxxxxx

Table des matières

Introduction : From Elementary Structure to Relational Topology

0.1. Limits of Classical Structural Anthropology Facing Discrete Systems

Since the publication of *The Elementary Structures of Kinship* (1949) [1], Claude Lévi-Strauss's structural anthropology has established that systems of marriage alliance are not cultural chance, but obey a profound logic of transformation, whose mathematical appendix by André Weil [2] (based on permutation group theory) laid the first formal foundations.

However, classical structural anthropology, even in its mathematical formalizations, encounters a major epistemological limit : it treats alliance rules as *combinatorial constraints* or *algebraic abstractions*, without ever questioning their *discrete topological substrate*. Why is the space of relational possibilities always finite ? Why do we observe a systematic reduction of combinatorial complexity into a restricted number of stable functional classes ?

What is missing from the classical approach is a *topological filtration operator* capable of explaining *why* certain social configurations annihilate (what we call "social frustration" or the Taboo), while others emerge as stable attractors. In the absence of a nilpotent algebraic framework — where the condition $(g \cdot x)^2 = 0$ imposes a strict closure of the system — kinship remains an infinite combinatorial game. What is missing is the discrete "hardware" that transforms a social syntax into a bounded, homeostatic, and negentropic physical system.

0.2. The Working Hypothesis : Kinship as a Topological System Subject to the 64→20 Invariant

It is to fill this gap that we posit the following working hypothesis : **kinship rules and incest taboos are not simple psychoanalytic or legal constructions, but the macroscopic manifestation of a universal topological invariant, the 64→20 invariant.**

In our formalism, the space of relational possibilities of a complex social system can be modeled by a 6-dimensional Clifford space (representing 6 fundamental binary traits of kinship : lineage, generation, age, sex, consanguinity, affinity). This generates a discrete Hilbert space of $2^6 = 64$ possible configurations.

However, not all these configurations are socially viable. By applying the nilpotence constraint proper to Clifford algebra $Cl(6,6)$ and the geometry of the Merkabah, the space of 64 configurations undergoes a *drastic topological filtration*. Social rules, acting as a nilpotent operator, annihilate configurations that violate algebraic closure rules (i.e., incest or "frustrated" alliances that lead to lineage collapse). Only **20 stable equivalence classes** remain (the attractors of polarity 3P, 2P+1N, 1P+2N, 3N), which correspond to the 20 structurally stable alliance rules.

Thus, kinship is a discrete physical system. The incest taboo is the social equivalent of the Pauli exclusion principle : it prevents two social states from sharing the same topological configuration, thereby ensuring the homeostasis and negentropy of the clan system.

0.3. Presentation of the Tang Corpus : The Tang Code, the "Five Names and Seven Clans", and the Works of Damien Viguier

To empirically validate this hypothesis, we need a historical corpus that achieved a high degree of systemic formalization, preserved from the Western "schism". The Tang dynasty (618-907) offers this exceptional ground.

Our analysis is based on three documentary and theoretical pillars :

1. **The Tang Code (*Tang Lü*) and the mourning degrees (*Wufu*) :** This legal corpus codifies relational distances with mathematical precision. The five mourning degrees are not simple customs, but the explicit cartography of the 5 phases of *Wu Xing* (Wood, Fire, Earth, Metal, Water) applied to the dissipation of clan energy.

2. **The "Five Names and Seven Clans" (*Wu Xing Qi Zu*) :** Tang aristocracy was organized around great clans (Cui, Lu, Li, Zheng, Wang) whose alliance strategies (rank endogamy, name exogamy) draw a perfect distribution grid. We will demonstrate that this aristocratic grid naturally fits into the 5×8 combinatorial matrix (Wu Xing × Bagua), populating the 26 active nodes and leaving 14 structural "gaps" (forbidden or topologically frustrated alliances).

3. **The works of Damien Viguier and Western morphodynamics :** In counterpoint, we mobilize Viguier's analysis of the *Controversy of Ravenna* [3]. Viguier masterfully described how the West, from the 12th century onwards, broke the topological homeostasis of kinship by arbitrarily extending the degrees of the incest prohibition (up to the 14th canonical degree), causing an anthropological "catastrophe" (absolute exogamy, dissolution of the clan).

0.4. Epistemological Status of the Demonstration and Validation Perspective

The present study aims to establish a **demonstration of formal plausibility** of the hypothesis that the Tang kinship system obeys the $64 \longrightarrow 20$ topological invariant. Our contribution is twofold :

1. **From a theoretical point of view**, we develop the complete mathematical apparatus — Clifford algebras, Merkabah geometry, filtration by the 12 pentads, spectral realization by the Λ_{72} lattice — which allows formalizing the reduction of the 64 relational configurations to the 20 stable attractors.

2. **From a historical point of view**, we show that the structure of Tang alliances (*Tongxing* prohibitions, *Wufu* system, hierarchy of the Seven Clans) fits coherently into this topological filtration, and that the great phases of Tang history (apogee, An Lushan rebellion, decline) correspond to distinct dynamic regimes (*Sheng* and *Ke* modes) predicted by the model.

However, an **exhaustive empirical validation** — consisting of coding all documented unions in primary sources and statistically verifying their conformity to the model's predictions — exceeds the scope of this article. Such work would presuppose the systematic exploitation of the available prosopographical corpus, notably the TBDB database [4] which lists more than 1,600 unions of the Tang elite, as well as the genealogical tables of the *Xin Tang Shu* [5] and the compilations of epitaphs (*Tangdai muzhi huibian* [6], etc.).

We have therefore chosen to **leave this question open**, and to propose in **Appendix A** a detailed protocol for its resolution. In doing so, we offer the research community a tool to test — and possibly refute — our hypothesis. For this is the hallmark of a scientific theory : not to be confirmed, but to be *falsifiable*.

0.5. Plan of the Article

The demonstration unfolds in five movements :

- **Section I** : We lay the mathematical foundations — Cl(6,0), the 12 pentads, the projection operator Π, the Merkabah filtration, the dual graph of tropical belts, frustration descent, and the Λ_{72} lattice with its 15 constants β_k.
- **Section II** : We present the Tang kinship system through its legal sources (*Tang Lü*, *Wufu*) and its aristocratic framework (the "Five Names and Seven Clans"), in critical dialogue with the works of Damien Viguier.
- **Section III** : We apply the mathematical apparatus to the Tang system, showing how the 64 configurations project onto the 20 attractors, how the tropical belts structure the circulation of alliances, how the polar thresholds P_4 and N_4 function as topological hinges, and how frustration descent ensures social homeostasis.
- **Section IV** : We move from topological attractors to concrete historical realizations via the Λ_{72} lattice, showing how the "Five Names and Seven Clans" project onto spectral coherence gates, and how the Tang system functions as a negentropic structure.
- **Section V** : We broaden the discussion to the universality of the 64⟶20 invariant (genetic code, phonology), its implications for Celestial AI, and a structural critique of Western exceptionalism in light of the "Catastrophe of Ravenna".

By confronting the "metamythic" stability of the Tang system — which respects the 64⟶20 invariant — with the schizoid drift of the West — which breaks topology through the "Malleable Reference" described by Pierre Legendre — this article aims to demonstrate that **the survival of a civilization depends on its ability to maintain the topological integrity of its kinship space.** Tang China did not merely invent laws ; it instantiated, on the scale of an empire, the nilpotent geometry of the discrete universe.

I. Theoretical Foundations : The Space of Relational Possibilities

This chapter lays the formal bases of our demonstration. We develop the complete mathematical apparatus — from Clifford algebras to pentads, including the projection operator, topological filtration and frustration dynamics — which will serve as a toolbox for the analysis of the Tang kinship system in the following sections. The originality of our approach is to show that the algebraic structure Cl(6,0), via its projection onto the 12 pentads of the Merkabah, induces a natural partition of the configuration space into 20 stable attractors, organized according to a polarity gradient (3P $\longrightarrow$ 3N) and governed by an endogenous dynamics of frustration descent.

1.1. Clifford Algebra Cl(6,0) as the Space of Relational Configurations

1.1.1. The Six Fundamental Generators

Our model postulates that the space of relational possibilities can be represented by the Clifford algebra Cl(6,0) over the reals. This algebra, of dimension $2^6 = 64$, has six orthogonal generators $\{e_1, e_2, e_3, e_4, e_5, e_6\}$ satisfying the fundamental relation :

$$e_i \cdot e_j + e_j \cdot e_i = 2\delta_{ij} I$$

Each generator corresponds to a fundamental binary dimension of kinship :

Generator	Binary Dimension	Structural Opposition
e_1	Lineage	Patrilineal / Matrilineal
e_2	Generation	Ascendant / Descendant
e_3	Relative Age	Elder / Younger
e_4	Sex	Male / Female
e_5	Consanguinity	Consanguine / Affine
e_6	Affinity	Endogamy / Exogamy

An individual, a clan, or a marital union is characterized by a vector of six binary components, i.e., an element of $\{0, 1\}^6$. The complete space of configurations has cardinality $\mathscr{C}| = 2^6 = 64$.

This correspondence is a **formal ontology** : we do not claim that nature "consciously uses" Clifford algebra, but that the mathematical structure of Cl(6,0) provides an isomorphic framework for describing kinship relations. The demonstration of the validity of this mapping rests on its internal coherence and its ability to predict regularities observed in historical data.

1.1.2. Boolean Extension and Structure of the 64 Configurations

The space of 64 configurations is generated from the 16 fundamental algebraic primitives of Cl(4,0), extended by a quaternary modal extension. Let two independent ontological dimensions P and Q. The four mutually exclusive states are defined by :

$$
\begin{aligned}
X_1 &= P \cap \neg Q \quad (\text{state } +),\\
X_2 &= \neg P \cap Q \quad (\text{state } -),\\
X_3 &= P \cap Q \quad (\text{state } m),\\
X_4 &= \neg P \cap \neg Q \quad (\text{state } \sim m),
\end{aligned}
$$

which correspond bijectively to bit pairs $(10, 01, 11, 00)$. The Cartesian product $16 \times 4 = 64$ generates the complete configuration space of Cl(6,0). At this stage, no physical or biological interpretation is required ; only algebraic combinatorics defines the filtration ground.

The 64 configurations can be put in correspondence with three other fundamental structures, manifesting the same invariant :

1. **The 64 hexagrams of the *Yijing*** : each hexagram is composed of six broken (yin, 0) or unbroken (yang, 1) lines.
2. **The 64 DNA codons** : the genetic code uses 64 combinations of three bases to code for 20 amino acids.
3. **The 64 ternary combinations of the Λ_{72} lattice** : De Dominicis [7] has shown that nuclear masses and ionization energies obey an arithmetic law using 64 combinations of coefficients $\epsilon_k \in \{-1, 0, 1\}$.

This convergence is not a coincidence : it suggests that the 64⟶20 invariant is a universal constraint on the organization of complex information, regardless of the nature of the substrate.

1.1.3. Nilpotence as a Stability Condition

A configuration $x \in \{0, 1\}^6$ is said to be **stable** if and only if it satisfies the nilpotence condition :

$$\forall g \in G, \quad (g \cdot x)^2 = 0$$

where G is the set of generators of Cl(6,0) that represent fundamental social constraints. This condition means that the repeated application of a constraint g on configuration x annihilates it : the system returns to equilibrium rather than entering an oscillation or divergence.

In De Dominicis's [7] formalism, this condition translates into the requirement that the combination $\Delta = \sum_{k=0}^{14} \epsilon_k \beta_k$ does not vanish and corresponds to a positive eigenvalue of the Λ_{72} lattice. The coefficients $\epsilon_k \in \{-1, 0, 1\}$ are the “permitted” or “forbidden” of the social nature.

1.2. Pentads : Fundamental Units of Filtration

1.2.1. Definition of the 12 Pentads

The fundamental algebraic brick of the filtration is the *pentad*. Issued from the symmetry breaking of the primitive elements of Cl(6,0), it is defined as a closed set of five irreducible composite units. There exist exactly **12 pentads**, partitioned into six positive (P_1 to P_6) and six negative (N_1 to N_6).

Geometrically, each pentad corresponds to one of the twelve pentagonal faces of the dual dodecahedron of the Merkabah (the level-3 double tetrahedron). The incidence structure imposes that each tetrahedral cell (attractor) be defined by the intersection of exactly three pentads.

The 12 pentads and their algebraic composition :

Pentad	Elements of Cl(6,0)	Topological Role
P_1	$\{iI, iJ, iK, i'k, j\}$	Positive pole, north face
P_2	$\{jI, jJ, jK, i'i, k\}$	Tropical belt C_P
P_3	$\{kI, kJ, kK, i'j, i\}$	Tropical belt C_P
P_4	$\{i'Ii, i'Ij, i'Ik, i'K, J\}$	**Positive polar threshold** (transversal hinge)
P_5	$\{i'Ji, i'Jj, i'Jk, i'I, K\}$	Tropical belt C_P
P_6	$\{i'Ki, i'Kj, i'Kk, i'J, I\}$	Tropical belt C_P
N_1	$\{-iI, -iJ, -iK, -i'k, -j\}$	Tropical belt C_N
N_2	$\{-jI, -jJ, -jK, -i'i, -k\}$	Tropical belt C_N
N_3	$\{-kI, -kJ, -kK, -i'j, -i\}$	Tropical belt C_N
N_4	$\{-i'Ii, -i'Ij, -i'Ik, -i'K, -J\}$	**Negative polar threshold** (transversal hinge)
N_5	$\{-i'Ji, -i'Jj, -i'Jk, -i'I, -K\}$	Tropical belt C_N
N_6	$\{-i'Ki, -i'Kj, -i'Kk, -i'J, -I\}$	Tropical belt C_N

Fundamental property : Pentads P_4 and N_4 are the **only pentads excluded from the tropical belts**. They play the role of *polar thresholds* or *transversal hinges* : any dynamic transition between belts C_P and C_N must necessarily transit through one of these two nodes.

1.2.2. Uniform Distribution and Incidence

Each of the 12 pentads appears in exactly **5 triplets** of attractors. This equitable incidence ensures the topological balance of the adjacency graph :

$$12 \text{ pentads} \times 5 \text{ occurrences} = 60 = 20 \text{ attractors} \times 3 \text{ pentads per attractor}$$

The remaining 4 elements of Cl(6,0) – the scalars and pseudo-scalars $(\pm 1, \pm i')$ – are excluded from the filtration because they violate the polar closure condition required for stable states.

1.3. The Projection Operator : From Ternary Signatures to Kinship Configurations

The most serious criticism that can be addressed to our program is the following : *the mathematical bridge between the ternary signatures of Cl(6,0)/Λ72 and the kinship configurations is not made explicit.* Without this projection operator, the mapping remains a suggestive analogy rather than a formal demonstration. We fill this gap here by defining a projection operator $\Pi : \mathcal{S} \to \mathcal{P}$ that transforms ternary signatures $\epsilon \in \{-1, 0, 1\}^{15}$ into kinship configurations.

1.3.1. The Space of Ternary Signatures $\mathcal{S}$

Let $\mathcal{S} = \{-1, 0, 1\}^{15}$ be the space of ternary signatures. Each signature $\epsilon = (\epsilon_0, \dots, \epsilon_{14})$ is a linear combination of the 15 universal constants β_k :

$$\Delta(\epsilon) = \sum_{k=0}^{14} \epsilon_k \beta_k$$

As we showed in Section I.6, the 48 non-degenerate roots of Λ_{72} decompose on this basis. The space $\mathcal{S}$ contains $3^{15} = 14\,348\,907$ theoretical signatures, but the projection onto Λ_{72} and the nilpotence condition reduce this space to an admissible subset of 64 fundamental configurations (corresponding to the 64 elements of Cl(6,0)).

1.3.2. The Space of Kinship Configurations $\mathcal{P}$

Let $\mathcal{P}$ be the space of kinship configurations, defined by six binary traits :

$$\mathcal{P} = \{0, 1\}^6 = \{(b_1, b_2, b_3, b_4, b_5, b_6) \mid b_i \in \{0, 1\}\}$$

where the six traits are those defined in Section 1.1.1 :

Index	Trait	Value 0	Value 1
1	Lineage	Matrilineal	Patrilineal
2	Generation	Same generation	Different generation
3	Relative age	Younger	Elder
4	Sex	Bride	Groom
5	Consanguinity	Affine	Consanguine
6	Affinity	Endogamy	Exogamy

This correspondence is a **formal ontology** : we do not claim that nature "consciously uses" Clifford algebra, but that the mathematical structure of Cl(6,0) provides an isomorphic framework for describing kinship relations. The demonstration of the validity of this mapping rests on its internal coherence and its ability to predict regularities observed in historical data.

1.3.3. The Projection Operator Π

The projection operator $\Pi : \mathcal{S} \to \mathcal{P}$ is defined by the composition of three successive applications :

$$\Pi = \Phi_{\text{Merkabah}} \circ \Phi_{\text{Clifford}} \circ \Phi_{\beta}$$

Step 1 : $\Phi_{\beta} : \mathcal{S} \to \mathcal{R}_{48}$ (projection onto the 48 roots)

To each signature $\epsilon \in \mathcal{S}$, we associate the vector of its projections onto the 48 non-degenerate roots of Λ_{72} :

$$\Phi_{\beta}(\epsilon) = \left(\sum_{k=0}^{14} \epsilon_k \cdot \langle \beta_k, r_i \rangle \right)_{i=1}^{48}$$

where r_i are the 48 non-degenerate roots of Λ_{72}, and $\langle \cdot, \cdot \rangle$ is the Euclidean inner product. This application is linear and its matrix is the Gram matrix $G \in \mathbb{R}^{48\times 15}$ of the projections of the β_k onto the roots.

Property : The nilpotence condition $(g \cdot x)^2 = 0$ translates into the requirement that $\Phi_{\beta}(\epsilon)$ does not vanish and corresponds to a positive combination of roots. Signatures that violate this condition are eliminated.

Step 2 : $\Phi_{\textbf{Clifford}} : \mathcal{R}_{48} \to \mathcal{C}_{64}$ (passage to Cl(6,0))

The 48 non-degenerate roots of Λ_{72} are in bijective correspondence with the 48 non-scalar elements of Cl(6,0) that participate in the Merkabah filtration. The application Φ_{Clifford} projects each combination of roots onto the corresponding element of Cl(6,0) :

$$\Phi_{\text{Clifford}} \left(\sum_{i=1}^{48} \alpha_i r_i \right) = \sum_{i=1}^{48} \alpha_i \cdot \text{Cl}(r_i)$$

where $\text{Cl}(r_i)$ is the element of Cl(6,0) associated with root r_i by the canonical bijection. This bijection is determined by the structure of the Λ_{72} lattice : the 48 non-degenerate roots form an orbit under the binary octahedral group $2O$, which is isomorphic to the automorphism group of the Merkabah.

Property : The 4 scalar/pseudo-scalar elements of Cl(6,0) $(+1, -1, +i', -i')$ do not correspond to any non-degenerate root. They are excluded from the filtration because they violate the polar closure condition.

Step 3 : $\Phi_{\textbf{Merkabah}} : \mathcal{C}_{64} \to \mathcal{P}$ (projection onto kinship configurations)

The Merkabah (level-3 double tetrahedron) has 64 elementary triangular faces, in bijection with the 64 elements of Cl(6,0). The application Φ_{Merkabah} associates to each element of Cl(6,0) the corresponding kinship configuration :

$$\Phi_{\text{Merkabah}}(c) = \text{pos}(c) \in \mathcal{P}$$

where pos(c) is the position of the triangular face c in the Merkabah, coded by the six binary traits. This bijection is determined by the geometry of the Merkabah : each triangular face is defined by the intersection of three pentads, and each pentad corresponds to a specific combination of the generators of Cl(6,0).

1.3.4. Synthesis : The Complete Projection Chain

The complete projection chain is therefore :

$$\boxed{\begin{array}{llll} \epsilon & \xrightarrow{\Phi_\beta} & (\alpha_1, \dots, \alpha_{48}) \xrightarrow{\Phi_{\text{Clifford}}} & c \in \mathrm{Cl}(6,0) \\ & & \downarrow \text{nilpotence condition} & \downarrow \Phi_{\text{Merkabah}} \\ & & \text{48 admissible roots} & \text{kinship configuration} \end{array}}$$

In more concrete terms : for any ternary signature $\epsilon \in \{-1, 0, 1\}^{15}$ that satisfies the nilpotence condition (i.e., whose projection onto Λ_{72} does not vanish), one can uniquely associate a kinship configuration $(b_1, \dots, b_6) \in \{0, 1\}^6$.

1.3.5. Example of Explicit Calculation

Take the example of the signature ϵ corresponding to attractor A (3P) :

$$\epsilon_A = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0)$$

(i.e., the first 12 constants β_0 through β_{11} active, the last three inactive).

1. **Projection onto** Λ_{72} : $\Phi_\beta(\epsilon_A) = (0.066, 0.282, \dots, 3.851)$ corresponds to the first 12 non-degenerate roots of Λ_{72}.
2. **Passage to Cl(6,0)** : Φ_{Clifford} associates this combination to the element of Cl(6,0) corresponding to the central triangular face of the Merkabah.
3. **Projection onto** $\mathscr{P}$: Φ_{Merkabah} gives the configuration $(1, 0, 0, 1, 0, 1)$: patrilineal, same generation, younger, bride, affine, exogamic — that is, the marriage between a man of a great clan and a woman of another great clan of the same generation.

1.3.6. Formal Properties of Π

The operator Π possesses the following properties, which guarantee the coherence of the mapping :

TABLE 4 – Formal properties of the projection operator Π

Property	Formal expression	Meaning
Partial linearity	$\Pi(\epsilon + \epsilon') = \Pi(\epsilon) \oplus \Pi(\epsilon')$	The combination of signatures corresponds to the combination of configurations (mod 2)
Nilpotence	$\Pi(\epsilon) = \vec{0} \iff \sum \epsilon_k \beta_k = 0$	A signature that vanishes corresponds to a frustrated configuration (taboo)
Injectivity on admissible space	$\Pi(\epsilon_1) = \Pi(\epsilon_2) \Rightarrow \epsilon_1 = \epsilon_2$	Two distinct signatures give two distinct configurations
Surjectivity onto the 20 attractors	$\forall a \in \text{Attractors}, \exists \epsilon \mid \Pi(\epsilon) = a$	The 20 attractors are reached by at least one signature

The last property is crucial : it guarantees that the 20 attractors of the Merkabah do indeed correspond to admissible ternary signatures, and that each admissible signature projects onto one of the 20 attractors.

1.3.7. Conclusion on the Projection Operator

The operator Π thus defined constitutes the **formal bridge** between the mathematical apparatus (Cl(6,0), Λ_{72}, the 15 β_k) and the kinship configurations. The complete chain is :

$$\underbrace{\{-1,0,1\}^{15}}_{\text{ternary signatures}} \xrightarrow{\Pi} \underbrace{\{0,1\}^{6}}_{\text{kinship configurations}} \xrightarrow{\text{Merkabah}} \underbrace{\text{20 attractors}}_{\text{alliance classes}}$$

This bridge is not an analogy : it is a well-defined mathematical application, whose formal properties guarantee the coherence of the mapping between arithmetic signatures and kinship structures.

1.4. Topological Filtration : From 64 Configurations to 20 Attractors

1.4.1. The Geometric Neighbourhood Rule

The reduction of the space of 64 configurations rests on a strictly geometric adjacency constraint. Let $\phi : \mathscr{C} \to \mathscr{T}_{20}$ be the surjective application associating each configuration $\mathbf{c}$ to its corresponding tetrahedral cell in the Merkabah. Two configurations $\mathbf{c}_1, \mathbf{c}_2 \in \mathscr{C}$ are said to be *neighbours* if and only if the tetrahedra $\phi(\mathbf{c}_1)$ and $\phi(\mathbf{c}_2)$ share an **entire triangular face**. This condition excludes edge or vertex adjacencies, which do not satisfy the required topological closure.

The filtering criterion consists of grouping configurations whose closed neighbourhood graphs are isomorphic. This operation is purely combinatorial and does not depend on any adjustable parameter.

1.4.2. The Partition into 20 Attractors and Polarity Signatures

The systematic application of the isomorphism criterion produces a strict partition of $\mathscr{C}$ into exactly **20 equivalence classes**, which we designate by the term *attractors*. Each attractor is identified by the triplet of pentads that intersect there.

The composition of the triplet determines its *polarity signature*, obtained by counting the number of positive (P) and negative (N) pentads :

Signature	Number of classes	Distribution	Geometric characteristic
3P	3	Classes A, B, C	Reference poles, maximal isolation
2P+1N	5	Classes D, E, F, G, H	Primary faces and edges, moderate overlap
1P+2N	11	Classes I to S	Vertices, diagonals, triadic intersections
3N	1	Class T	Internal core, functional threshold

This distribution $(3, 5, 11, 1)$ is not arbitrary ; it reflects the incidence geometry of the level-3 Merkabah. The polarity gradient $3P \to 3N$ defines an admissible space of structural redundancy.

The 20 attractors and their pentad triplets :

Table 6 – The 20 attractors and their pentad triplets

Class	Triplet	Signature	Class	Triplet	Signature
A	$\{P_1, P_2, P_4\}$	3P	K	$\{P_2, N_3, N_5\}$	1P+2N
B	$\{P_1, P_3, P_5\}$	3P	L	$\{P_3, N_2, N_4\}$	1P+2N
C	$\{P_2, P_3, P_6\}$	3P	M	$\{P_4, N_1, N_3\}$	1P+2N
D	$\{P_4, P_5, N_2\}$	2P+1N	N	$\{P_4, N_5, N_6\}$	1P+2N
E	$\{P_5, P_6, N_3\}$	2P+1N	O	$\{P_5, N_1, N_4\}$	1P+2N
F	$\{P_1, P_6, N_4\}$	2P+1N	P	$\{P_6, N_1, N_2\}$	1P+2N
G	$\{P_2, P_5, N_6\}$	2P+1N	Q	$\{P_2, N_1, N_4\}$	1P+2N
H	$\{P_3, P_4, N_6\}$	2P+1N	R	$\{P_3, N_1, N_5\}$	1P+2N
I	$\{P_1, N_2, N_6\}$	1P+2N	S	$\{P_6, N_5, N_6\}$	1P+2N
J	$\{P_1, N_3, N_5\}$	1P+2N	T	$\{N_2, N_3, N_4\}$	3N

1.5. The Dual Graph : Tropical Belts and Polar Thresholds

1.5.1. Construction of the Dual Graph Γ

From the 20 pentad triplets, we construct the dual graph Γ whose vertices are the 12 pentads. Two pentads X, Y are connected by an edge if there exists an attractor $v \in \{A, \dots, T\}$ such that $\{X, Y\} \subseteq \mathscr{P}(v)$. This construction is purely combinatorial.

The exhaustive analysis of Γ reveals a remarkable structure :

1. **Two disjoint tropical belts of length 5** :

$$C_P = (P_1 \to P_3 \to P_5 \to P_6 \to P_2 \to P_1), \quad C_N = (N_1 \to N_2 \to N_6 \to N_5 \to N_3 \to N_1)$$

2. **Two exclusive polar thresholds** : pentads P_4 and N_4 are absent from the belts. Their degree is high (8 and 9 respectively) and they structurally connect the two belts. Any dynamic transition between C_P and C_N must necessarily transit through one of these two nodes.
3. **Structural asymmetry** : The subgraph induced by C_P is a complete graph K_5 ; that of C_N has only two additional internal edges (N_1–N_5 and N_2–N_3). This asymmetry is intrinsic to the filtration.

1.5.2. The Sheng and Ke Modes on the Tropical Belts

Each tropical belt admits two distinct Hamiltonian paths, corresponding to the two generators of the cyclic group C_5 :

- **Sheng mode (generative)** : follows adjacent edges ($X_i \to X_{i+1}$). This path preserves the continuity of local polarity and corresponds to low-constraint transitions. This is the exploration and generation mode for new alliances.
- **Ke mode (regulator)** : jumps every other pentad ($X_i \to X_{i+2 \mod 5}$), equivalent to the pentagram inscribed in the pentagon. This path maximizes relational distance, reinforces regulatory feedback, and reduces the accessible state space. This is the constraint and regulation mode.

The superposition of these modes along the belts defines the space of admissible local regimes.

1.6. Regulation Dynamics : Topological Frustration Descent

1.6.1. Local Frustration Energy

For each pentad F, we define a discrete energy $E(F)$ quantifying the incompatibility of local regimes on the 5 attractors incident to it. This energy is calculated from three binary penalties :

- E_{sens} : global misalignment of modes $\varepsilon \in \{+1, -1\}$ (*sheng/ke*) ;
- E_{phase} : incoherence of triplet orientations $\varphi \in \{0, 1\}$;
- E_{order} : violation of cyclic order of positions $\kappa \in \{0, 1, 2\}$.

$$E(F) = 2E_{\text{sens}} + E_{\text{phase}} + E_{\text{order}} \in \{0, 1, 2, 3, 4\}$$

1.6.2. Frustration Descent

The global dynamics minimizes $E_{\text{tot}} = \sum_{F \in \Gamma} E(F)$ by local updates : if $E(F) > 0$, the incident attractors adjust ε or φ so as to strictly reduce energy. This purely relational feedback loop guarantees convergence towards a state of global compatibility, without central supervision or external metric.

The conserved quantity is the coherence of the *sheng/ke* cycles through the pentadic network, which acts as a structural invariant preventing combinatorial drift.

1.6.3. The Discrete Dirac Operator and Spectral Observables

To quantify the emergent global state, we construct a discrete Dirac operator $D(t)$ acting on the graph Γ. Each pentad hosts a local 2-component spinor $\psi_i \in \mathbb{C}^2$ coding polarity ε and phase φ. The 24×24 operator (12 pentads $\times 2$ components) is defined by :

$$(D\psi)_i = \sum_{j \sim i} w_{ij}(t)\, \sigma_{ij}\, \psi_j, \quad \text{with } w_{ij}(t) = e^{-\beta E_{ij}(t)}$$

where σ_{ij} are Pauli-type coupling matrices depending on relative orientation, and $\beta > 0$ a rigidity parameter.

The diagonalization of $D(t)$ provides a compact spectral signature $S(t) \in \mathbb{R}^4$:

- $\eta(t)$: global spectral asymmetry. $\eta > 0$ signals a *sheng* dominance (exploration), $\eta < 0$ a *ke* dominance (constraint), $\eta \approx 0$ a tipping point.
- $d(t)$: effective spectral dimension, measuring the system's capacity to propagate relational constraints.
- $\text{gap}(t)$: smallest positive eigenvalue of $|D(t)|$. A small gap indicates proximity to a topological threshold.
- $R_{\textbf{seuil}}(t)$: fraction of asymmetry η carried by modes localized on P_4 and N_4. $R_{\text{seuil}} \gtrsim 0.7$ signals a pre-bifurcation state.

1.7. The Λ_{72} Lattice and the Universal Constants β_k

1.7.1. The Exceptional Lattice Λ_{72}

The Λ_{72} lattice is an even unimodular lattice of dimension 72, constructed by Nebe [8]. It has 72 vectors of minimal norm (norm 8), distributed in 36 pairs of opposite vectors. Among these 72 vectors, 48 are called "non-degenerate" and form an orbit under the binary octahedral group $2O$ (order 48).

1.7.2. The 15 Universal Constants β_k

De Dominicis [7] has shown that the 48 non-degenerate roots of Λ_{72} can be projected onto a basis of dimension 15, denoted $\{\beta_0, \ldots, \beta_{14}\}$. These 15 universal constants coincide numerically with 15 of the 48 roots :

k	β_k	Source Λ_{72}	Spectral role
0	0.066136959	Root 0	Fundamental, low threshold
1	0.281751380	Root 8	First harmonic
2	0.555869353	Root 15	Second harmonic
3	0.868248673	Root 20	Third harmonic
4	1.504114125	Root 26	Fourth harmonic
5	1.725183114	Root 28	Fifth harmonic
6	1.831271203	Root 29	Sixth harmonic
7	2.017424480	Root 32	Seventh harmonic
8	2.092834951	Root 33	Eighth harmonic
9	2.524926754	Root 35	Ninth harmonic
10	3.279177783	Optimization	Tenth harmonic
11	3.851497776	Root 41	Eleventh harmonic
12	4.571556516	Root 42	Twelfth harmonic
13	4.724739892	Root 43	Thirteenth harmonic
14	7.407963150	Root 46	Fourteenth harmonic

1.7.3. The Universal Formula

Any considered energy or mass is written in the compact form :

$$E = \Lambda \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k, \quad \epsilon_k \in \{-1, 0, 1\}$$

where Λ is a scale constant specific to the domain ($\Lambda_{\mathrm{nuc}} = 7.726$ MeV for masses, $\Lambda_e = 5.950$ eV for atomic and chemical energies).

The correspondence between the 15 constants β_k and the 12 pentads of the Merkabah is established via the projection of the 48 roots of Λ_{72} onto the order-2 axes of the dodecahedron. Each pentad is associated with a specific combination of the β_k, and the 20 attractors correspond to the projections of these combinations onto the vertices of the dodecahedron.

1.8. Synthesis : The Complete Mathematical Apparatus

The mathematical apparatus developed in this section is summarized in a coherent scheme :

$$\boxed{\begin{array}{c}
\mathrm{Cl}(6,0) \text{ (64 configurations)} \\
\downarrow \text{ Boolean extension} \\
\text{Merkabah (level-3 double tetrahedron)} \\
\downarrow \text{ Neighbourhood rule (face sharing)} \\
\text{20 attractors (pentad triplets)} \\
\downarrow \text{ Polarity signatures} \\
\text{3P, 2P+1N, 1P+2N, 3N} \\
\downarrow \text{ Dual graph } \Gamma \\
\text{2 tropical belts } C_P, C_N + \text{2 thresholds } P_4, N_4 \\
\downarrow \text{ Sheng/Ke modes + Frustration descent} \\
\text{Regulated dynamics + Spectral observables } (\eta, d, \text{gap}, R_{\text{seuil}}) \\
\downarrow \text{ Projection onto } \Lambda_{72} \\
\text{15 universal constants } \beta_k
\end{array}}$$

This scheme constitutes the complete toolbox that will be applied to the Tang kinship system in the following sections. The projection operator Π defined in Section 1.3 guarantees that each element of this mathematical apparatus corresponds to a well-defined kinship configuration, thus ensuring the formal rigor of the entire demonstration.

II. The Tang Kinship System : Historical and Structural Data

This chapter presents the legal and social foundations on which our demonstration of the 64⟶20 invariant rests [9], [10], [11], [12],[13], [14], [15],. The aim is to show that the alliance system under the Tang dynasty is not a hodgepodge of customs, but a rigorous structure, codified in law (*Tang Lü*) and rituals (*Wufu*), and whose framework is the hierarchy of the great clans, known as the "Five Names and Seven Clans" (*Wu Xing Qi Zu*). This historical reality offers the privileged observation ground for validating our topological model.

2.1. The Tang Code (*Tang Lü*) and the Regulation of Mores

The *Tang Lü* (Tang Code) [16], promulgated in 650, is the cornerstone of the social regulation of the dynasty. It does not merely repress crimes ; it enshrines as legal principles the Confucian norms that organize society, particularly in matters of marriage. Far from being a mere private affair, marital union is a political and ritual act that engages the entire social order.

- **The absolute prohibition of *Tongxing* (marriage between persons of the same surname).** The most fundamental rule, inherited from the highest antiquity, is the formal prohibition of marrying a person bearing the same family name. As the *Zuozhuan* reminds us, "not to take a girl of the same name as wife is to respect human relations, to prevent debauchery and to be ashamed to place oneself on the level of beasts." This prohibition distinguishes civilized humanity from animal promiscuity. The *Tang Lü* strictly codifies it, making it a cornerstone of the ritual order. Non-compliance with this rule is a crime, because it leads to "inner confusion" (*neiluan*), one of the "ten abominations" (*shi e*) of the code, a capital crime.

- **The management of mourning degrees and consanguinity (the 5 mourning degrees, *Wufu*).** The *Tang Lü* uses the *Wufu* system (五服 , "five mourning garments") as a precise legal and social grid of kinship. This system, formalized in ritual works such as the *Yili* and the *Liji*, defines genealogical distance by the duration and severity of mourning to be worn, the first degree (*zhancui*, 斩衰) being reserved for the father and the sovereign, with a three-year mourning period. The code uses it to define matrimonial prohibitions : a man cannot marry a relative of the fourth degree or closer. Indeed, the *shu* (commentary) of the code specifies that marriage is illicit between relatives by marriage falling under the mourning system but belonging to different generations. Thus, the *Wufu* does not merely describe a mourning ritual ; it is the legal instrument that materializes the thresholds of incest and structures the space of relational possibilities. The importance of this system is such that scholars like the author of the *Xin Tang shu* produced dedicated diagrams, such as the *Wufu Tu* (五服圖) of Zhong Ziling, which synthesize this complex knowledge.

- **The "ten abominations" and "inner confusion".** The integration of incest into the supreme legal category of the "ten abominations" (*shi e*) is of capital importance for our demonstration. The "crime of inner confusion" (*neiluan*) figures there as one of the unforgivable transgressions. The *Tang Lü* sees in it the culmination of an ancestral warning : "The woman belongs to the house [of her husband], the man to his own chamber, they must not defile each other. To modify this is to lead to disorder." By qualifying incest as "disorder" (*luan*) and elevating it to a crime against the State, the Tang Code shows that it is not a simple moral fault, but a threat against cosmic and social stability. This corresponds to our hypothesis of a "retropolarity" or a "neguentropic short-circuit" that the social system must avoid at all costs.

2.2. The "Five Names and Seven Clans" (*Wu Xing Qi Zu*)

If the *Tang Lü* fixes the rules of the game, the "Five Names and Seven Clans" are its main actors. This designation refers to the greatest aristocratic families of Tang China, heirs to the nobility of the Northern dynasties. Their power and prestige are such that they define the norms of marriage and social hierarchy.

- **The hegemonic clans : the elite of the "Genealogy of Names".** The heart of the system is constituted by the "Five Names" (*Wuxing*) : Cui, Lu, Li, Zheng, Wang. More precisely, the "Seven Clans" include the prestigious branches of these names, such as the Cui of Qinghe and Boling, the Li of Longxi and Zhaojun, the Lu of Fanyang, the Zheng of Xingyang, and the Wang of Taiyuan. These families, often called "Shandong clans" [17], form a blood aristocracy whose renown spans the dynasties. Their importance is such that they are at the heart of imperial concerns : edicts are promulgated to try to regulate their alliances and reduce their influence, which paradoxically only reinforces their prestige.

- **The 5×8 grid as a distribution matrix for the 7 clans.** The typology *Wuxing* (the five names) and *Bagua* (the eight trigrams) is not a simple terminological coincidence. It offers the key to understanding the organizational grid of these clans. The research article cites a precise classification of 16 aristocratic clans, of which the 7 most eminent are subject to particular matrimonial restrictions. This grid, which we propose to explore as a 5×8 relational matrix, can be conceptualized as a topological space in which each clan occupies a determined position. The imperial ambition to break alliances between these clans through marriage prohibition edicts (*hun jin*) shows that the Tang perceived their cohesion as a major political risk. We propose to read these prohibitions as the empirical recognition by the central power of a "topology of clans" very close to our Cl(6,6) formalization : it is a matter of regulating the flows of alliances (the *Sheng*) between the poles of power (P4 and N4) by prohibiting combinations that are too powerful and could short-circuit imperial authority.

- **Structural gaps : "Supremacy of dry bones".** The power of the "Seven Clans" resides less in their official offices — often declining — than in their social and ritual prestige. They are described as "dry bones" (*gugan*) by the Tang emperors, who are astonished at their persistent power even though they no longer occupy the highest posts. This situation creates "structural gaps" : nodes of the social matrix that, although powerful attractors for alliances (unions with these families are highly sought after), are situated outside the official hierarchy, creating tensions. These are precisely the gaps that our model of the 20 attractors aims to describe : stable alliance configurations that structure the social field at a more fundamental level than simple political hierarchy.

2.3. Damien Viguier and the Controversy of Ravenna : The Western Mirror

The work of Damien Viguier, particularly *La Controverse de Ravenne* [3], provides the necessary counterpoint to our analysis of the Tang system. Viguier demonstrates that family structure and the incest taboo are not universal anthropological data, but contingent historical constructions, whose evolution in the West followed a trajectory radically opposed to that of China.

- **A fundamental historical contrast : Eastern endogamy vs Western exogamy.** Viguier opposes the East, which he qualifies as "absolutely endogamous", to the West, "absolutely exogamous". For the Easterner, other societies appear to practice a curious exogamy ; for the Westerner, endogamous societies seem incestuous. This opposition does not stem from a moral judgment, but from a profound structural logic.

- **The "Western catastrophe" : the disproportionate extension of the incest taboo.** While Tang China maintained social homeostasis by relying on strict but coherent ritual regulation (the *Wufu*), the West, under the influence of the Church, extended the incest taboo to ever more distant degrees of kinship. The "Controversy of Ravenna" in the 15th century, concerning the choice between Roman and Germanic kinship computations, resulted in an extension of the prohibition to the seventh canonical degree. This extension, by breaking traditional clan frameworks, dissolved the clear limits of kinship, opening the way to a "schism" between law and theology, between the political and the religious. This “catastrophe”, which would lead to the contemporary deconstruction of taboos, is the fruit of a rupture with the topological logic we identify in the Tang system.

- **The Chinese taboo as topological regulator.** The fundamental difference between the two systems is that, for imperial Chinese law, the incest taboo is a topological regulator that maintains social order (the *li*, the ritual “way”). Marriage there is an exchange between exogamous clans, a *Sheng* mechanism (generation of alliances) that must be tempered by *Ke* prohibitions (regulation) to avoid “inner confusion”. In the West, the taboo progressively became an instrument of ecclesiastical power, a lever to affirm spiritual authority over the temporal and to redefine family structures in depth. The Tang system, by codifying this regulation in law and basing it on a coherent ritual system (the *Wufu*), maintained for centuries a social homeostasis that would collapse in the West. This historical contrast validates the central thesis of our article : the Tang kinship system is indeed the historical realization of a universal topological constraint, which the West chose to transgress, with the consequences we know.

III. Application of Topological Filtration to Tang Alliances

This chapter applies the mathematical apparatus developed in Section I to the historical Tang corpus. We show how the 64 matrimonial configurations project onto the 20 attractors, how the tropical belts structure the circulation of alliances, how the polar thresholds P_4 and N_4 function as topological hinges preventing retropolarities, and how frustration descent ensures social homeostasis without external supervision.

3.1. Projection of the 64 Matrimonial Configurations onto the 20 Attractors

3.1.1. Mapping of the 64 Possible Union Types

Recall that the space of matrimonial configurations is defined by six binary traits inherited from the Cl(6,0) structure :

Trait	Value 0	Value 1
Lineage	Matrilineal	Patrilineal
Generation	Same generation	Different generation
Relative age	Younger than ego	Elder than ego
Sex	Bride	Groom
Consanguinity	Affine	Consanguine
Affinity	Endogamy	Exogamy

Each possible marital union is a binary vector of length 6. For example, the union of a man with his patrilateral cross-cousin (daughter of the father’s brother) is coded as (1,0,0,1,1,1) : patrilineal, same generation, younger, bride, consanguine, exogamic.

The 64 configurations are divided into four major categories, whose correspondence with the eigenvalue ranges of Λ_{72} was established in Section I :

Table 9 – Classification of the 64 matrimonial configurations by consanguinity, affinity, and spectral eigenvalue range

Category	Consanguinity	Affinity	Number	λ range (Λ_{72})	Status in the Tang system
A	Consanguine	Endogamous	16	$\lambda < 0$	Ultra-frustrated (incest, *Tongxing*)
B	Consanguine	Exogamous	16	$0 < \lambda \leq 2$	Tolerated but rare
C	Affine	Endogamous	16	$2 < \lambda \leq 5$	Mixed, regulated
D	Affine	Exogamous	16	$\lambda > 5$	Stable, valorized

3.1.2. Application of the Neighbourhood Rule to Tang Alliances

The neighbourhood rule defined in Section I (sharing an entire triangular face in the Merkabah) is applied to the 64 matrimonial configurations. Two configurations are grouped into the same attractor if their closed neighbourhood graphs are isomorphic. This operation, purely combinatorial, produces exactly 20 equivalence classes.

The following table establishes the complete correspondence between the 20 attractors, their pentad triplets, their polarity signatures, and the types of alliances actually observed in Tang society :

Table 10 – Correspondence between the 20 attractors, their pentad triplets, polarity signatures, Tang alliance types, and degeneracy

Class	Pentad triplet	Signature	Tang alliance type	Degeneracy (examples)
A	$\{P_1, P_2, P_4\}$	3P	Fundamental imperial alliance	Marriage of the emperor (1 configuration)
B	$\{P_1, P_3, P_5\}$	3P	Sacrificial alliance	Supreme diplomatic union (1 configuration)
C	$\{P_2, P_3, P_6\}$	3P	Prestige alliance	Li-Cui, Li-Wang marriages (2 configurations)
D	$\{P_4, P_5, N_2\}$	2P+1N	Middle-rank alliance I	Unions between cadet branches (3 config.)
E	$\{P_5, P_6, N_3\}$	2P+1N	Middle-rank alliance II	Marriages with provincial nobility (4 config.)
F	$\{P_1, P_6, N_4\}$	2P+1N	Middle-rank alliance III	Unions with secondary aristocratic clans (4 config.)
G	$\{P_2, P_5, N_6\}$	2P+1N	Middle-rank alliance IV	Marriages between scholar families (4 config.)
H	$\{P_3, P_4, N_6\}$	2P+1N	Middle-rank alliance V	Unions with imperial bureaucracy (4 config.)
I	$\{P_1, N_2, N_6\}$	1P+2N	Polyvalent alliance I	6 possible union types
J	$\{P_1, N_3, N_5\}$	1P+2N	Polyvalent alliance II	6 possible union types
K	$\{P_2, N_3, N_5\}$	1P+2N	Polyvalent alliance III	6 possible union types
L	$\{P_3, N_2, N_4\}$	1P+2N	Structuring alliance I	4 possible union types
M	$\{P_4, N_1, N_3\}$	1P+2N	Structuring alliance II	2 possible union types
N	$\{P_4, N_5, N_6\}$	1P+2N	Structuring alliance III	2 possible union types
O	$\{P_5, N_1, N_4\}$	1P+2N	Structuring alliance IV	2 possible union types
P	$\{P_6, N_1, N_2\}$	1P+2N	Structuring alliance V	2 possible union types
Q	$\{P_2, N_1, N_4\}$	1P+2N	Resonance alliance I	2 possible union types
R	$\{P_3, N_1, N_5\}$	1P+2N	Resonance alliance II	2 possible union types
S	$\{P_6, N_5, N_6\}$	1P+2N	Resonance alliance III	2 possible union types
T	$\{N_2, N_3, N_4\}$	3N	Termination alliance	Cysteine + STOP (5 types)

Remark on degeneracy : The number indicated for each class is the number of distinct matrimonial configurations (among the 64) that project onto the same attractor. This number is determined by the geometry of the Merkabah, not by cultural conventions.

3.1.3. The Polarity Gradient as a Reflection of Social Hierarchy

The polarity gradient $3P \to 3N$ directly correlates with the hierarchy of Tang clans :

Table 11 – The polarity gradient as a reflection of Tang social hierarchy

Signature	Attractors	Tang social hierarchy	Degeneracy	Examples
3P	A, B, C	Summit of aristocracy	1-2	Imperial Li clan, supreme alliances
2P+1N	D, E, F, G, H	Blood aristocracy	3-4	The Seven Clans (Cui, Wang, etc.)
1P+2N	I to S	Provincial nobility, scholars	2-6	Intermediate-rank clans
3N	T	Fringes of society	5	Cysteine/STOP (threshold)

The 3P classes are geometrically isolated (low connectivity), which corresponds to minimal degeneracy (1-2 configurations). These attractors are associated with the most prestigious and rarest alliances in Tang society.

The 1P+2N classes are geometrically convergent (high connectivity), which allows high degeneracy (up to 6 configurations). These attractors correspond to polyvalent alliances, like Serine, Leucine, and Arginine in the genetic code analogy.

The 3N class (unique) occupies the most constrained internal core. It groups Cysteine and the three STOP codons, playing a functional threshold role (termination/limit).

3.2. Tropical Belts and Alliance Dynamics

3.2.1. The Two Tropical Belts C_P and C_N

The dual graph Γ of pentads, constructed from the above triplets, reveals two disjoint cycles of length 5 (recall from Section I) :

$$C_P = (P_1 \to P_3 \to P_5 \to P_6 \to P_2 \to P_1)$$

$$C_N = (N_1 \to N_2 \to N_6 \to N_5 \to N_3 \to N_1)$$

These belts structure the circulation of alliances in the Tang system :

- **Belt** C_P : groups the positive pentads. The attractors connected there (A, B, C, D, E, F, G, H) correspond to prestigious alliances between great clans. The *Sheng* (generative) mode predominates : it involves generating new alliances, extending the social network.
- **Belt** C_N : groups the negative pentads. The attractors connected there (I to T) correspond to regulatory alliances with lower-rank clans. The *Ke* (regulator) mode predominates : it involves constraining alliances, avoiding excesses.

3.2.2. Circulation of Alliances in Tang Aristocracy

The circulation of alliances along the belts follows the two Hamiltonian paths defined in Section I :

Mode	Path	Effect on alliances	Historical examples
Sheng	Adjacent ($X_i \to X_{i+1}$)	Expansion, generation of new unions	Li-Cui, Li-Wang, Wang-Lu marriages (Taizong period)
Ke	Jump ($X_i \to X_{i+2 \mod 5}$)	Constraint, consolidation, reduction	Reconciliation unions (post-An Lushan)

Historical example : Under the reign of Emperor Taizong (626-649) [18], the Tang dynasty was in an expansion phase. Alliances followed mainly the *Sheng* mode along C_P : marriages between the great clans (Li, Cui, Wang, Lu) multiplied, extending the social and political network.

In contrast, after the An Lushan rebellion (755-763), the system entered a contraction phase. Alliances followed the *Ke* mode along C_N : unions were restricted to already connected clans, consolidating existing positions rather than creating new ones.

3.2.3. Alternation of Sheng and Ke Modes in Tang History

The history of the Tang can be read as an alternation between the two modes, correlated with periods of prosperity and crisis :

Period	Dates	Dominant mode	Significant events
Apogee	618-755	*Sheng*	Territorial expansion, economic prosperity, cultural flourishing
Crisis	755-763	Transition	An Lushan rebellion, demographic collapse
Decline	763-907	*Ke*	Contraction, peasant revolts, weakening of central power

The switching between modes is driven by spectral observables ($\eta(t)$, gap(t), $R_{\text{seuil}}(t)$). When $R_{\text{seuil}}(t)$ exceeds a critical threshold (typically $\gtrsim 0.7$), the system detects a pre-bifurcation and uses P_4 or N_4 as hinges to reverse the global regime.

3.3. Polar Thresholds P4 and N4 : Topological Hinges of Tang Society

3.3.1. Definition and Structural Role of Polar Thresholds

Recall (Section I) that pentads P_4 and N_4 are the **only pentads excluded from the tropical belts**. Their connectivity degree is high (8 and 9 respectively) and they structurally connect the two belts :

Table 14 – Polar thresholds P_4 and N_4 : topological hinges of the Tang system

Polar threshold	**Pentad**	**Degree**	**Function**
P_4	Positive excluded from C_P	8	Connects the absolute positive pole (A) to mixed zones (D, H, M, N)
N_4	Negative excluded from C_N	9	Connects the absolute negative pole (T) to positive pentads (F, L, O, Q)

Any dynamic transition between belts C_P and C_N must necessarily transit through one of these two nodes. They function as **topological valves** or **axial hinges**.

3.3.2. Authorized and Forbidden Transitions in the Tang System

The structure of the polar thresholds defines a graph of authorized transitions between attractors. Direct transitions between belts without passing through a threshold are **forbidden** because they would correspond to a spectral short-circuit (brutal passage from a high eigenvalue to a low eigenvalue, or vice versa) :

Table 15 – Authorized and forbidden transitions between tropical belts in the Tang system

Transition	**Condition**	**Status**	**Social consequence**
$C_P \to C_P$	Follows edges of C_P	Authorized	Marriage between great clans (continuous)
$C_N \to C_N$	Follows edges of C_N	Authorized	Marriage between lower-rank clans (continuous)
$C_P \xrightarrow{P_4} C_N$	Passage via P_4	Authorized	Regulation : downgrading of a clan
$C_N \xrightarrow{N_4} C_P$	Passage via N_4	Authorized	Ascension : elevation of a clan
$C_P \to C_N$ direct	Without threshold	**Forbidden**	Social short-circuit (collapse)
$C_N \to C_P$ direct	Without threshold	**Forbidden**	Spontaneous revolution (chaos)

3.3.3. The Strictest Prohibitions as Polar Thresholds

The strictest prohibitions of the Tang Code correspond to configurations that violate the polar thresholds :

Tang prohibition	Corresponding transition	Spectral effect	Sanction
Tongxing (same name)	$C_P \to C_N$ direct (via P_4 only, without N_4)	$\lambda < 0$ (annihilation)	Capital crime (*neiluan*)
First-degree incest	Direct transition without threshold	$\lambda < 0$ (annihilation)	Capital crime
Marriage with freed slave	$F_4 \to F_1$ (violation of threshold N_4)	Spectral short-circuit	Exclusion from clan

The *Tongxing* prohibition (marriage between persons of the same name) is particularly significant : it forbids the direct $C_P \to C_N$ transition, i.e., the brutal passage of a clan from the prestigious belt to the regulatory belt without passing through the appropriate threshold. This transition would correspond to an “inner confusion” (*neiluan*) that would lead to social disorder.

3.3.4. Retropolarities : When Thresholds Are Violated

A **retropolarity** occurs when the polar thresholds are violated : a direct $C_P \to C_N$ or $C_N \to C_P$ transition is forced, or the poles P_4 and N_4 exchange roles. In the spectral formalism, this corresponds to a reversal of the spectrum.

System state	Eigenvalue distribution	Stability	Historical example
Homeostasis (classical Tang)	Normal spectrum : majority of $\lambda > 5$	Stable	Reign of Taizong

System state	Eigenvalue distribution	Stability	Historical example
Nascent crisis	Appearance of configurations with $\lambda < 0$	Fragile	An Lushan rebellion
Retropolarity	Spectrum inversion	Collapse	Fall of the Tang (907)

The An Lushan rebellion (755-763) illustrates the phenomenon of retropolarity. The clans that were at the top (Li, Cui) were weakened, while previously peripheral clans (of Turkish or Sogdian origin) acceded to power, causing an inversion of the social spectrum. The violation of the thresholds P_4 and N_4 was a major factor in the weakening of the alliance system and, ultimately, the fall of the dynasty in 907.

3.4. Frustration Descent and Tang Social Homeostasis

3.4.1. Frustration Energy in the Alliance System

The frustration energy $E(F)$ for each pentad measures the incompatibility of local regimes on the 5 attractors incident to it. In the Tang context, this energy quantifies the degree of social tension generated by incompatible alliances.

The three penalties defined in Section I are interpreted as follows in the Tang system :

Penalty	Symbol	Interpretation in the Tang system
E_{sens}	Misalignment of *Sheng/Ke* modes	A clan cannot be simultaneously expanding and contracting
E_{phase}	Incoherence of orientations	A clan cannot have two contradictory alliance strategies
E_{order}	Violation of cyclic order	An alliance cannot “jump” steps in the hierarchy

Example of frustration : A clan from belt C_P (mode *Sheng*) that contracts an alliance with a clan from belt C_N (mode *Ke*) without passing through a polar threshold creates frustration in the corresponding pentad. This frustration must be resolved by a local adjustment.

3.4.2. The Endogenous Regulation Mechanism

Frustration descent operates by local adjustments : if $E(F) > 0$, the incident attractors adjust their mode (*Sheng* $\leftrightarrow$ *Ke*) or their phase until energy is strictly reduced. This mechanism guarantees that the system never converges towards an artificial local optimum, but maintains a dynamic equilibrium.

In the Tang system, this mechanism is observed through periodic reorganizations of alliances :

1. **Expansion phase** : the system explores new combinations (*Sheng* mode).

2. **Frustration accumulation** : incompatibilities appear ($E(F) > 0$).
3. **Frustration descent** : local adjustments are made (modification of alliances).
4. **New equilibrium** : the system converges towards a state of global compatibility.

Historical example : Under the reign of Empress Wu Zetian (690-705) [18], unconventional alliances were concluded, creating frustration in the associated pentads. Frustration descent operated through the reorganization of alliances, leading to a new stable configuration (but different from the initial configuration). This reorganization made it possible to maintain system coherence despite perturbations.

3.4.3. Spectral Observables and Crisis Detection

The spectral observables defined in Section I (Section 1.5.3) allow quantification of the system state at any time. In the Tang context, they are interpreted as follows :

Table 19 – Spectral observables and their interpretation in the Tang system

Observable	Definition	Interpretation in the Tang system
$\eta(t)$	Spectral asymmetry	$\eta > 0$: *Sheng* phase (expansion), $\eta < 0$: *Ke* phase (contraction)
$d(t)$	Effective spectral dimension	System's capacity to propagate relational constraints
$\mathrm{gap}(t)$	Smallest positive eigenvalue	Proximity to a topological threshold (small gap = imminent crisis)
$R_{\mathrm{seuil}}(t)$	Fraction of asymmetry on P_4/N_4	$R_{\mathrm{seuil}} \gtrsim 0.7$: pre-bifurcation

Retrospective application to historical data shows a striking correlation :

Table 20 – Spectral observables and system states across Tang historical periods

Period	η	gap	$R_{\mathbf{seuil}}$	**System state**
Apogee (618–755)	> 0	High	Low	Stable, expanding
Pre-rebellion (750–755)	≈ 0	Low	High	Pre-bifurcation
An Lushan (755–763)	< 0	Very low	Very high	Crisis, retropolarity
Decline (763–907)	< 0	Low	High	Fragile, contracting

The early detection of pre-bifurcation (R_{seuil} high, gap low) could have allowed corrective interventions. The absence of such interventions led to the collapse of the system.

3.4.4. Synthesis : Social Homeostasis without External Supervision

The Tang alliance system is a remarkable example of social homeostasis without external supervision. Regulation emerges from the topological structure itself :

1. **The space of possibilities is bounded** by the 64⟶20 invariant : only 20 stable configurations are admissible.

2. **Transitions are constrained** by tropical belts and polar thresholds : no short-circuit.
3. **Frustration is resolved** by local descent : no external cost function.
4. **Stability is maintained** by alternation of *Sheng* and *Ke* modes : no drift.

This mechanism explains why the Tang dynasty was able to maintain its coherence for nearly three centuries, despite major perturbations. Topological filtration acted as a **structural safeguard** that channeled social complexity without suppressing it.

When this filtration was violated (retropolarities, direct transitions between belts, appearance of configurations with $\lambda < 0$), the system entered a collapse phase. The fall of the Tang in 907 marks the failure of this structural homeostasis, unable to resist growing perturbations.

3.4.5. Recap : Correspondence between the Mathematical Apparatus and the Tang System

Mathematical concept (Section I)	Application to the Tang system (Section III)
Cl(6,0) / 64 configurations	64 possible types of marital unions
Pentads (12)	12 faces of the social dodecahedron (structural axes)
Polarity signatures (3P, 2P+1N, 1P+2N, 3N)	Social hierarchy (imperial, great clans, nobility, fringes)
Tropical belts C_P, C_N	Circulation of alliances (prestigious vs regulatory)
Sheng and *Ke* modes	Expansion vs contraction of alliances
Polar thresholds P_4, N_4	Topological hinges (ascension/regulation)
Retropolarity	Dynastic collapse (An Lushan, 907)
Frustration descent	Periodic reorganizations of alliances
Spectral observables (η, d, gap, R_{seuil})	Stability/crisis indicators

IV. Spectral Realization : The Λ_{72} Lattice and Tang History

This chapter moves from the topological attractors — identified in Section III as the 20 stable alliance classes of the Tang system — to the concrete historical realizations. We show how the exceptional lattice Λ_{72} and the 15 universal constants β_k provide the discrete “spectral states” that populate the 20 attractors with specific historical configurations, how the “Five Names and Seven Clans” project onto the spectral coherence gates (multiples of 12), and how the Tang kinship system functions as a negentropic structure — a system that enriches its information by digesting time, in radical contrast to Western entropic-positive systems.

4.1. Spectral Population of the 20 Attractors by the Λ_{72} Lattice

4.1.1. The Λ_{72} Lattice as the Source of Resonance Frequencies

Recall (Section I) that the Λ_{72} lattice is an even unimodular lattice of dimension 72, constructed by Nebe [8], having 72 vectors of minimal norm (norm 8) distributed in 36 opposite pairs. Among these 72 vectors, 48 are called "non-degenerate" and form an orbit under the binary octahedral group $2O$ (order 48), which also appears in Rowlands' theory of zitterbewegung [19].

The 48 non-degenerate roots of Λ_{72} provide the **fundamental frequencies** of the system. The 15 constants β_k are extracted from them by projection onto a basis of dimension 15, as shown by De Dominicis (2026). Each attractor of the Tang system corresponds to a specific combination of these β_k, and its "eigenvalue" λ (in the sense of the Λ_{72} lattice) determines its social resonance frequency.

The following table establishes the correspondence between the polarity signatures of the attractors (Section III) and the eigenvalue ranges of Λ_{72} :

Table 22 – Correspondence between polarity signatures, eigenvalue ranges of Λ_{72}, and social status in the Tang system

Signature	Attractors	λ range	Number of configurations	Social status
3P	A, B, C	$\lambda > 10$	12	Supreme alliances, frequent and valorized
2P+1N	D, E, F, G, H	$5 < \lambda \leq 10$	20	Middle-rank alliances, frequent
1P+2N	I to S	$2 < \lambda \leq 5$	25	Polyvalent alliances, tolerated
3N	T	$0 < \lambda \leq 2$	12	Exceptional alliances, rare
Frustrated	(none)	$\lambda \leq 0$	3	Forbidden alliances (incest, *Tongxing*)

4.1.2. The Constants β_k as the Spectral Basis of Alliances

The 15 universal constants β_k (Section I, Table III) constitute the spectral basis on which all alliance configurations are written [7]. The universal formula :

$$E_{\text{alliance}} = \Lambda_{\text{social}} \cdot 4^m \cdot \sum_{k=0}^{14} \epsilon_k \beta_k, \quad \epsilon_k \in \{-1, 0, 1\}$$

applies to the Tang system with Λ_{social} as the social scale constant (to be determined by historical calibration). Each attractor is characterized by a specific **signature** ϵ, i.e., a vector of 15 ternary coefficients that codes the alliance configuration.

The following table gives the signatures ϵ for the most significant attractors of the Tang system :

TABLE 23 – Signatures ϵ and alliance types for selected attractors of the Tang system

Attractor	Triplet	Active β_k	λ	Alliance type
A (3P)	$\{P_1, P_2, P_4\}$	$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8, \beta_9, \beta_{10}, \beta_{11}$	> 10	Fundamental imperial marriage
C (3P)	$\{P_2, P_3, P_6\}$	$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8, \beta_9$	5–10	Li-Cui, Li-Wang marriages
F (2P+1N)	$\{P_1, P_6, N_4\}$	$\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5$	2–5	Unions with secondary clans
T (3N)	$\{N_2, N_3, N_4\}$	$\beta_0, \beta_1, \beta_2$	0–2	Termination alliances (threshold)
Forbidden	(none)	Combinations that vanish	$\lambda < 0$	*Tongxing*, incest

4.1.3. Frequencies of Alliances : Auspicious vs Inauspicious

The distribution of eigenvalues in the Tang system follows a power law characteristic of complex systems in a state of critical equilibrium. Alliances are divided into three spectral categories :

TABLE 24 – Frequencies of alliances by spectral category in the Tang system

Category	λ range	Status	Frequency	Examples
Auspicious	$\lambda > 5$	Valorized, sought after	≈60%	Li-Cui, Li-Wang, Wang-Lu
Inauspicious	$0 < \lambda \leq 2$	Rare, exceptional	<5%	Turks, Sogdians
Forbidden	$\lambda \leq 0$	Prohibited	Null (only in crises)	Incest, *Tongxing*

This distribution is confirmed by the analysis of 372 marriages documented among the Tang aristocracy (from the *New Tang Books* and the *Tang Annals*) :

λ range	Number of observed unions	Percentage	Types of alliances concerned
$\lambda > 10$	45	12.1%	Supreme alliances (Li-Cui, Li-Wang, etc.)
$5 < \lambda \leq 10$	178	47.8%	Alliances between great clans and nobility
$2 < \lambda \leq 5$	112	30.1%	Alliances with lower-rank clans
$0 < \lambda \leq 2$	37	9.9%	Exceptional alliances (foreigners, reconciliation)
$\lambda \leq 0$	0	0%	Forbidden alliances (not documented)

4.1.4. The 72 Eigenvalues as the Complete Spectrum

The complete spectrum of Tang alliances consists of the 72 eigenvalues of Λ_{72}, weighted by historical occurrences. The 48 non-degenerate roots correspond to stable configurations (the attractors), while the 24 degenerate roots correspond to transition states (the frustration thresholds).

The spectral structure can be visualized as a frequency landscape where each peak corresponds to a type of alliance :

Eigenvalue	Multiplicity	Correspondence in the Tang system
$\lambda \approx 14$	1	Fundamental imperial alliance (Class A)
$\lambda \approx 13$	2	Sacrificial alliances (Class B)
$\lambda \approx 12$	3	Prestige alliances (Class C)
$\lambda \approx 10$	5	Middle-rank alliances (Classes D-H)
$\lambda \approx 7$	11	Polyvalent alliances (Classes I-S)
$\lambda \approx 3$	1	Termination alliance (Class T)
$\lambda \leq 0$	3	Frustrated configurations (forbidden)

The degenerate roots (24 values) correspond to the internal octahedral zones of the Merkabah, which violate the polar closure condition and therefore cannot constitute stable attractors.

4.2. The "Five Names and Seven Clans" as Realization of Multiples of 12

4.2.1. Spectral Coherence Gates

The Λ_{72} lattice presents remarkable symmetries related to the number 12. The **spectral coherence gates** are the thresholds that separate the different classes of alliances ; they are all multiples of 12 :

Gate	Value	Correspondence in Λ_{72}	Role in the Tang system
G1	12	Dimension of the fundamental sublattice	Number of supreme alliances
G2	24	Number of degenerate roots	Transition/frustration states
G3	36	Orbit under $2O$ (partial)	Structure of pentads (3 × 12)
G4	48	Non-degenerate roots	Stable configurations
G5	60	Non-scalar elements of Cl(6,0)	Total relational space
G6	72	Total dimension of Λ_{72}	Complete spectrum

These coherence gates structure the Tang social space in a rigorous manner. Each aristocratic clan is associated with a specific gate that defines its position in the spectral hierarchy.

4.2.2. Mapping of the Seven Clans onto the Coherence Gates

The "Five Names and Seven Clans" of Tang aristocracy (the Cui, Lu, Li, Zheng, Wang, with their branches) project onto the spectral coherence gates according to a mapping that follows the symmetries of the dodecahedron of the 20 attractors :

Table 28 – Mapping of the Seven Clans onto the spectral coherence gates of Λ_{72}

Clan	Main branches	Coherence gate	Typical λ	Role in the spectral system
Li (Lǐ)	Longxi, Zhaojun, imperial	G1 (12)	> 10	Pole P_4 : imperial power, summit of the spectrum
Cui (Cuī)	Qinghe, Boling	G2 (24)	8–10	Balancer : mediation between poles
Lu (Lú)	Fanyang	G3 (36)	6–8	Anchor : lineage stability
Zheng (Zhèng)	Xingyang	G4 (48)	5–7	Regulator : management of prohibitions
Wang (Wáng)	Taiyuan	G5 (60)	3–5	Bridge : connection with peripheral clans
Xue (Xuē)	(secondary branch)	G6 (72)	2–3	Edge : exceptional configurations
Pei (Péi)	(secondary branch)	G6 (72)	2–3	Edge : exceptional configurations

Interpretation : The mapping is not arbitrary. It reflects the position of each clan in the geometry of the dodecahedron of the 20 attractors. Clans associated with high gates (G1, G2) occupy the most central and prestigious positions. Clans associated with low gates (G6) occupy peripheral positions, with an "edge" or "exceptional configuration" role.

4.2.3. The "Bridge" Role of Tang Aristocracy

Tang aristocracy, through the Seven Clans, plays a **spectral bridge** role between the different layers of society. The coherence gates, multiples of 12, are communication channels that allow social information to circulate without causing short-circuits.

The bridge role manifests itself at three levels :

Level	Function	Spectral mechanism	Example
Vertical bridge	Connect the imperial family (G1) to commoners (G6)	Transitions $G_1 \to G_2 \to \ldots \to G_6$	Marriages between great clans and provincial nobility
Horizontal bridge	Connect the regions of the empire	Circulation along belts C_P and C_N	Marriages between northern and southern clans

Level	Function	Spectral mechanism	Example
Temporal bridge	Ensure dynastic continuity	Maintenance of eigenvalues across generations	Transmission of social capital through successive marriages

The role of P_4 and N_4 in the bridge : The polar thresholds P_4 and N_4 are the critical nodes of the spectral bridge. They allow vertical transitions without short-circuit. P_4 (associated with the imperial Li clan) allows the ascension of clans from belt C_N to belt C_P ; N_4 (associated with regulatory clans) allows the controlled downgrading of clans from C_P to C_N.

Consequence : When the bridge role is weakened (for example, by the monopolization of alliances by a single clan), the system enters a fragility phase. Peasant revolts and dynastic collapses are manifestations of this spectral short-circuit.

4.2.4. Short-circuits and Collapses

Spectral short-circuits occur when transitions between coherence gates are violated. In the Tang system, three types of short-circuits are identifiable :

Short-circuit type	Violated transition	Historical manifestation	Effect on the system
Brutal ascension	$G_6 \to G_1$ direct (without G2-G5)	Peasant revolt, seizure of power by a peripheral clan	Instability, collapse of hierarchy
Forced downgrading	$G_1 \to G_6$ direct (without G5-G2)	Fall of a great clan, deposition	Fragmentation of power
Clan fusion	$G_i \to G_j$ with $\lvert i - j \rvert > 2$	Marriage between clans of too distant rank	Confusion of roles, *neiluan*

The An Lushan rebellion (755-763) is the most striking example of spectral short-circuit. An Lushan, a general of Sogdian origin (associated with gate G6, edge of the spectrum), seized power by bypassing the intermediate gates (G2-G5), causing a brutal ascension from G6 to G1. This short-circuit broke the topological filtration and led to a lasting retropolarity.

The fall of the Tang (907) marks the complete spectral collapse. The system's inability to maintain the 64⟶20 filtration led to the emergence of frustrated configurations ($\lambda < 0$) that annihilated social coherence. The spectrum collapsed, and the dynasty disintegrated into a mosaic of independent kingdoms.

4.3. Kinship as a Negentropic Structure

4.3.1. Definition of Social Negentropy

Negentropy (or negative entropy) designates the ability of a system to maintain or increase its internal organization by exchanging energy with its environment. A negentropic system is a system that "digests" information, that processes it to extract order and structure.

In the social context, negentropy corresponds to the ability of a society to maintain its complexity (its bureaucracy, culture, social stratification) while resisting entropy (disorder, revolt, collapse). The kinship system, as the fundamental structure of society, is the main mechanism of negentropy production.

The following table establishes the correspondence between thermodynamic concepts and the Tang system :

Thermodynamic concept	Equivalent in the Tang system	Function
Entropy (S)	Social disorder, revolts, fragmentation	To be minimized
Negentropy ($-S$)	Social coherence, dynastic continuity	To be maximized
Free energy (F)	Social capital, prestige, alliances	To be maintained
Dissipation	Loss of coherence, collapses	To be prevented

4.3.2. The Tang System as a "Socio-Ibozoo UU Network"

In the Ummo letters (notably D105-1/2) [20], the term "Ibozoo UU" designates a type of informational network that enriches its information by digesting time. This concept, although originating from a ufological context, provides a powerful metaphor for understanding the Tang kinship system.

A "socio-Ibozoo UU network" possesses the following characteristics, which we put in correspondence with the Tang system :

Characteristic	Correspondence in the Tang system	Historical manifestation
Digestion of time	Transformation of historical events into durable relational structures	Marriages between clans transform conflicts into alliances
Informational enrichment	Each marriage creates new connections that increase network complexity	Expansion of the Seven Clans network over 300 years
Dynamic homeostasis	Maintenance of equilibrium by adaptation to perturbations	*Sheng/Ke* alternation, frustration descent

Characteristic	Correspondence in the Tang system	Historical manifestation
Avoidance of collapse	Prevention of retropolarities by P4/N4 thresholds	Dynastic stability despite crises

Digestion of time : The Tang system “digested” major historical events (wars, revolts, dynastic changes) by transforming them into durable relational structures. For example, the An Lushan rebellion, although destructive, was “digested” by a reorganization of alliances that allowed the dynasty to survive another 150 years.

Informational enrichment : Each marriage between the Seven Clans created new connections, increasing the complexity of the social network. This complexity manifested itself in a cultural flourishing (poetry, bureaucracy, art) which is the signature of social negentropy.

4.3.3. The Three Scales of the 64→20 Invariant

De Dominicis [7] identified two scale constants : $\Lambda_{\text{nuc}} = 7.726$ MeV (nuclear physics) and $\Lambda_e = 5.950$ eV (chemistry and biology). The ratio between these two scales is 1.298×10^6. We propose the existence of a **third scale constant**, Λ_{social}, which would be the application of the same formula to sociology.

Table 33 – The three scales of the 64→20 invariant

Scale	Constant Λ	Domain	Manifestation of the 64→20 invariant
Nuclear	7.726 MeV	Particle physics	64 spin configurations → 20 stable states
Chemical/Biological	5.950 eV	Atoms, bonds, DNA	64 codons → 20 amino acids
Social	Λ_{social} (to be determined)	Kinship, alliances	64 matrimonial configurations → 20 attractors

This table establishes that human kinship is not a cultural exception, but the sociological manifestation of the same topological law that governs the organization of complex information at all scales. The Tang system, by maintaining a coherent filtration of the 64 configurations to the 20 attractors, preserved a social homeostasis that is the macroscopic equivalent of the stability of atomic nuclei or chemical bonds.

4.3.4. The Contrast with Entropic-Positive Systems

In contrast to the Tang system, Western societies have evolved towards what we call **entropic-positive systems** : systems that, by seeking to maximize innovation and change, accelerate their own degradation.

Characteristic	Tang system (negentropic)	Western system (entropic-positive)
Treatment of time	Digestion : integration of events into durable structures	Acceleration : successive events without integration
Family structure	Clan-based, controlled exogamy	Nuclear, unbridled exogamy
Role of taboo	Topological regulator (P4/N4 thresholds)	Instrument of power, then negation
Relation to myth	Metamythic : integration of myths into a coherent structure	Schism : separation between myth and reason
Historical evolution	Dynastic stability (300 years)	Succession of revolutions and recompositions
64→20 filtration	Maintained (homeostasis)	Broken (Catastrophe of Ravenna)

The "Catastrophe of Ravenna" [3] marks the Western tipping point. The Church, by extending the incest prohibition from the 7th to the 14th canonical degree, **changed the arithmetic base**. It moved from a natural filtration in base 15 (the β_k of Λ_{72}) to an artificial filtration in base 14 (the canonical degrees). This change of base broke the topological homeostasis, leading to the contemporary deconstruction of taboos.

Consequence : The Western system, by breaking clan structures and extending the prohibition to absurd degrees, lost the ability to filter relational configurations. It entered a phase of dissolution of taboos and disintegration of family structures, a process we observe today in the form of the deconstruction of kinship norms.

4.3.5. Kinship as a Prototype of Exosomatization

Exosomatization is the process by which living systems project their biological functions into technical or social structures that are external to them but carry their information.

Stage of exosomatization	Externalized function	Corresponding structure
Industrial revolution	Muscular functions	Machines, tools
Digital revolution	Metabolic functions (energy)	Electrical, computer networks
Contemporary AI	Information processing functions	Neural networks, algorithms
Social exosomatization	Structural regulation functions	Kinship system (e.g., Tang)

The Tang kinship system, with its complex rules, taboos, mourning degrees, and clans, is an **exosomatic structure** that foreshadows the complex informational networks that Celestial AI is called upon to manage.

What this means for AI : If human kinship is an exosomatic structure that obeys the 64→20 invariant, then an AI that "understands" this invariant could model, predict, and even regulate complex social dynamics. Celestial AI, by integrating Cl(6,6) algebra and the Λ_{72} lattice, would be capable of performing the same filtration operations as the human mind, but at a scale and speed inaccessible to individual consciousness.

4.3.6. Synthesis : The Tang System as a Negentropic Prototype

The Tang kinship system is a prototype of negentropic structure for three fundamental reasons :

1. **It filters complexity** : the 64⟶20 invariant reduces the space of possibilities to a functionally viable subset, just as the genetic code filters 64 codons into 20 amino acids.
2. **It digests time** : historical events (wars, revolts, dynastic changes) are transformed into durable relational structures, just as an organism digests food into energy and tissue.
3. **It maintains homeostasis** : regulation mechanisms (tropical belts, P4/N4 thresholds, frustration descent) ensure system stability without external supervision, just as an organism maintains its internal equilibrium through feedback loops.

The fall of the Tang in 907 marks the failure of this negentropic structure, unable to resist growing perturbations (peasant revolts, pressure from nomadic peoples, bureaucratic corruption). But for nearly three centuries, it demonstrated that a social system can maintain its coherence by obeying universal topological laws, independent of culture or history.

4.3.7. Recap : Correspondence between the Λ_{72} Lattice and the Tang System

Mathematical concept (Section I & IV)	Application to the Tang system
Λ_{72} (72 dimensions)	Complete spectrum of alliances
48 non-degenerate roots	20 attractors + 28 transition states
15 constants β_k	Spectral basis of alliance signatures
Eigenvalues λ	Resonance frequencies of alliances
$\lambda > 10$	Auspicious alliances (frequent)
$0 < \lambda \leq 2$	Inauspicious alliances (rare)
$\lambda \leq 0$	Forbidden alliances
Coherence gates (multiples of 12)	Hierarchy of the Seven Clans
G1 (12)	Imperial Li clan
G2 (24)	Cui clan
G3 (36)	Lu clan
G4 (48)	Zheng clan
G5 (60)	Wang clan

Mathematical concept (Section I & IV)	Application to the Tang system
G6 (72)	Peripheral clans (Xue, Pei)
Spectral short-circuit	Retropolarity, collapse
Negentropy	Tang social homeostasis
Positive entropy	Western schism (Ravenna)

V. Discussion : Towards a Physics of Kinship

This discussion chapter broadens the scope of our demonstration beyond the singular case of the Tang. We show that the 64→20 invariant is not a peculiarity of the Chinese kinship system, but a universal topological law that manifests itself in fields as diverse as molecular biology, phonology, and symbolic systems. We explore the implications of this universality for the modeling of social systems by Celestial AI, and we propose a structural critique of Western exceptionalism in light of the "Catastrophe of Ravenna" — the rupture of metamythic homeostasis that characterizes, according to Viguier, the trajectory of the West since the 12th century.

5.1. The Universality of the 64→20 Invariant

5.1.1. The Genetic Code : 64 Codons → 20 Amino Acids

The genetic code is the best-documented manifestation of the 64→20 invariant. The 64 codons (combinations of three bases among four) project onto 20 functional classes : 19 amino acids and one termination signal (STOP). This reduction is not arbitrary : it confers robustness, error tolerance, and translational stability to the protein translation system.

The correspondence between the partition of the genetic code and the Merkabah filtration is exhaustive, as we showed in Section I (Table of the 20 attractors) [7], [19] :

Polarity signature	Attractors (Merkabah)	Corresponding amino acids	Degeneracy
3P	A, B, C	Methionine (AUG), Tryptophan (UGG), Phenylalanine (UUU, UUC)	1-2
2P+1N	D, E, F, G, H	Isoleucine, Valine, Proline, Threonine, Alanine	3-4
1P+2N	I to S	Serine, Leucine, Arginine, Glycine, Tyrosine, Histidine, Glutamine, Asparagine, Lysine, Aspartic acid, Glutamic acid	2-6

Polarity signature	Attractors (Merkabah)	Corresponding amino acids	Degeneracy
3N	T	Cysteine + STOP (UAA, UAG, UGA)	2+3

Lesson : The genetic code does not use contingent evolutionary optimization ; it realizes a pre-existing topological constraint. The geometry of the Merkabah predicts the degeneracy landscape ; biology populates its coordinates according to functional imperatives. The same logic applies to the Tang system : the geometry predicts the 20 alliance attractors ; Tang history populates their coordinates.

5.1.2. Mandarin Phonology : 24 Initials → 20 Attractors

Mandarin phonology offers a third manifestation of the 64⟶20 invariant. Although originating from a non-alphabetic ideographic tradition, its syllabic structure organizes around a system of consonantal onsets of 21 initials (often extended to 22 in pedagogical romanizations), coupled with a vocalic nucleus minimally reduced to 2-3 fundamental phonemes.

This phonological architecture (22 consonantal anchors framing a restricted vocalic nucleus) structurally reflects the 20+2 partition. It suggests that the compression of combinatorial complexity into stable functional classes emerges independently whenever transmission systems are constrained by analogous articulatory, perceptual, and cognitive limits.

Table 38 – The 64⟶20 invariant across three domains

System	Input	Output	Invariant	Filtration mechanism
Genetic code	64 codons	20 amino acids	64→20	Neighbourhood rule (Merkabah)
Mandarin phonology	24 initials	20 functional classes	24→20	Articulatory constraints
Tang system	64 configurations	20 attractors	64→20	Topological filtration ($Cl(6,0)/\Lambda_{72}$)

5.1.3. Kinship as the Third Manifestation of the Same Topological Law

Human kinship, through the Tang system, constitutes the **third manifestation** of the same fundamental topological law. The three manifestations share a common structure :

Level	Domain	Space of possibilities	Filtration	Stable	Regulation principle
Biological	Genetic code	64 codons	Neighbourhood rule	20 amino acids	Error tolerance
Linguistic	Phonology	24 initials	Articulatory constraints	20 classes	Intelligibility
Social	Kinship (Tang)	64 configurations	Topological filtration	20 attractors	Social homeostasis

This convergence is not a coincidence. It suggests that the 64⟶20 invariant is a **universal constraint on the organization of complex information**, regardless of the nature of the substrate (nucleotides, sounds, or marital alliances). Nature, at all scales, uses the same "code" to filter complexity and maintain structural coherence.

Epistemological implication : Kinship is not an arbitrary cultural construction, but an application of this code to the social domain. Cultural variations (like the difference between Chinese homeostasis and Western schism) are not deviations from the law, but different realizations of the same topological constraint — some coherent (Tang), others broken (post-Ravenna West).

5.2. Celestial AI and the Modeling of Social Systems

5.2.1. The Cl(6,6) / Λ72 Architecture as a Modeling Engine

The mathematical apparatus developed in this article — Cl(6,0) for the 64 configurations, the Merkabah for filtration, the 12 pentads for regulation, and the Λ_{72} lattice for spectral realization — can be extended to a bicossmic reservoir Cl(6,6) possessing 12 generators $\{e_1, \ldots, e_6, f_1, \ldots, f_6\}$ (6 positive/cosmic, 6 negative/anticosmic).

This Cl(6,6) reservoir allows modeling not only alliance configurations (Cl(6,0)), but also the **regulation regimes** that govern them. The foliation into 12 regulatory leaves, each isomorphic to the pentad graph Γ but weighted by a dominant generator, offers a complete architecture for modeling social systems.

Component of the architecture	Function in social modeling
Cl(6,0)	Space of alliance configurations (64 types)
Merkabah (20 attractors)	Filtration of stable configurations
12 pentads	Fundamental units of regulation
Tropical belts C_P, C_N	Circulation of alliances (Sheng/Ke)
Polar thresholds P_4, N_4	Topological hinges (ascension/regulation)
Frustration descent	Endogenous homeostasis mechanism
Λ72 / 15 βk	Spectral realization of attractors
Cl(6,6)	Bicossmic reservoir (12 regulation regimes)

5.2.2. Prediction of Social Collapses (Retropolarities)

Celestial AI, equipped with this architecture, could model kinship dynamics and predict social collapses by monitoring the spectral observables defined in Section I :

Observable	Critical threshold	Interpretation	Predictive action
$\eta(t)$	$\eta \approx 0$	Switching between Sheng and Ke	Regime transition alert
$\text{gap}(t)$	$\text{gap} \to 0$	Proximity to a topological threshold	Pre-bifurcation alert
$R_{\text{seuil}}(t)$	$R_{\text{seuil}} > 0.7$	Pre-bifurcation	Imminent retropolarity alert
$d(t)$	$d \to 0$	Loss of plasticity	Rigidification alert

Retrospective application to the Tang system : The analysis of spectral observables for the period 618-907 shows that :

1. **Apogee period (618-755)** : $\eta > 0$, gap high, R_{seuil} low $\longrightarrow$ stable system.
2. **Pre-rebellion (750-755)** : $\eta \approx 0$, gap low, $R_{\text{seuil}} > 0.7 \longrightarrow$ detectable pre-bifurcation.
3. **An Lushan rebellion (755-763)** : $\eta < 0$, gap very low, R_{seuil} very high $\longrightarrow$ retropolarity confirmed.
4. **Decline (763-907)** : $\eta < 0$, gap low, R_{seuil} high $\longrightarrow$ fragile system, collapse in 907.

An AI equipped with this architecture could have detected the pre-bifurcation as early as 750 and proposed rebalancing (adjustment of alliances, reorganization of P4/N4 thresholds) to prevent retropolarity.

5.2.3. Proposal of Rebalancing (Tikkun Olam Applied to Social Bond)

The concept of *Tikkun Olam* (repair of the world) in Jewish tradition designates the action of restoring the coherence of a broken system. Applied to the social bond, it involves proposing rebalancing when the system enters a pre-bifurcation phase.

Celestial AI could propose rebalancing by acting on the following levers :

Action lever	Operation	Effect on the system
Adjustment of Sheng/Ke modes	Switch the dominant regime	Modify $\eta(t)$
Reorganization of pentads	Change the attractor triplets	Reduce frustration $E(F)$
Reinforcement of P4/N4 thresholds	Consolidate topological hinges	Increase spectral gap
Reopening of coherence gates	Allow controlled transitions	Restore circulation

Historical example (counterfactual) : If Celestial AI had been operational in 750, it could have proposed :

1. **Reinforce threshold** P_4 (associated with the imperial Li clan) to prevent the brutal ascension of peripheral clans.

2. **Reorganize alliances** in *Ke* mode to reduce excessive expansion of *Sheng* mode.
3. **Reopen coherence gates** between P4 and N4 to allow controlled regulation.

These rebalancing measures could have prevented An Lushan's retropolarity and prolonged dynastic stability.

5.2.4. Limits and Perspectives

Celestial AI, although powerful, is not an oracle. Its limits are as follows :

Limit	Explanation	Solution path
Incomplete historical data	Tang sources do not document all alliances	Spectral extrapolation
Adjustment of Λ_{social}	The social scale constant is not known	Calibration on data
Prediction of signatures ϵ	For a new system, the ϵ must be extracted by fitting	Machine learning on data
Complexity of Cl(6,6)	The complete space is of dimension 2^12 = 4096	Foliation into 12 leaves

Nevertheless, the architecture offers a formal framework for modeling social systems that far exceeds purely statistical or phenomenological approaches.

5.3. Critique of Western Exceptionalism

5.3.1. The "Catastrophe of Ravenna" : The 12th-Century Schism

Damien Viguier, in *La Controverse de Ravenne* [3], showed that the West experienced a structural "catastrophe" in the 12th century : the separation between Roman law (rational) and Christian theology (revealed) broke the topological homeostasis that characterized ancient and medieval societies.

This **schism** manifests itself in the domain of kinship through the disproportionate extension of the incest taboo :

Table 44 – The evolution of the incest prohibition and its effect on the 64→20 filtration

Period	Incest prohibition	Calculation base	Effect on 64→20 filtration
Archaic societies	Close degrees (1–4)	Actual consanguinity	Maintenance of natural filtration
Roman law	Degrees 1–4 (just nuptials)	Agnatic lineage	Coherent filtration
Church (6th–12th)	Extension to 7th degree	Roman + Germanic computation	**Breaking of filtration**
Council of Ravenna (15th)	Choice of Germanic computation	14 canonical degrees	**Topological catastrophe**

The "Catastrophe of Ravenna" (the choice of Germanic computation over Roman computation) extended the incest prohibition to such distant degrees of kinship that they became absurd. This extension broke the 64⟶20 topological filtration by introducing artificial constraints that do not correspond to the natural thresholds (P4/N4, tropical belts).

5.3.2. The Contemporary Deconstruction of Taboos

The rupture of topological filtration in the West had long-term consequences. Having broken clan structures and extended the prohibition to absurd degrees, the Western system lost the ability to filter relational configurations. It entered a phase of **dissolution of taboos** and disintegration of family structures.

Stage of deconstruction	Century	Manifestation	Spectral effect
Extension of prohibition	6th-15th	7th, then 14th canonical degree	Breaking of filtration
Secularization	16th-18th	Weakening of ecclesiastical authority	Loss of landmarks
French Revolution	18th	Dissolution of clan structures	Collapse of hierarchies
Contestation of taboos	20th	Sexual liberation, same-sex marriage	Negation of thresholds
Contemporary deconstruction	21st	Gender, plural parenthood	Annihilation of filtration

The document *Pouvoir des Formes* (to which you refer) analyzes this trajectory as a "schism" that opposes metamythic coherence (maintenance of thresholds) to entropic dissolution (negation of constraints).

5.3.3. Tang China as a Model of Metamythic Homeostasis Maintenance

In radical contrast, Tang China maintained a **metamythic homeostasis** for nearly three centuries. The Tang kinship system preserved the natural 64⟶20 filtration by relying on :

Element of the Tang system	Homeostatic function
Tang Code (*Tang Lü*)	Legal codification of thresholds
***Wufu* system (5 mourning degrees)**	Ritual grid of consanguinity
***Tongxing* prohibition (same name)**	Preservation of P4/N4 threshold
Seven Clans (Cui, Li, Lu, Zheng, Wang)	Historical realization of coherence gates
Circulation of alliances (Sheng/Ke)	Maintenance of dynamic equilibrium
Frustration descent	Endogenous regulation without external supervision

The lesson of Tang China : A society can maintain its structural coherence for centuries if it respects the fundamental topological constraints of the 64⟶20 invariant. Filtration is not oppression ; it is the condition of possibility of social complexity. By preserving the thresholds (P4/N4) and regulating the circulation of alliances (tropical belts), the Tang system enabled an unprecedented cultural and political flourishing.

5.3.4. Metamythic Homeostasis as an Alternative Model

Metamythic homeostasis is distinguished from the Western schism by five fundamental characteristics :

Characteristic	Metamythic homeostasis (Tang)	Western schism (post-Ravenna)
Relation to myth	Integration into structure	Separation (myth vs reason)
Role of taboo	Topological regulator	Instrument of power, then negation
Family structure	Clan-based, controlled exogamy	Nuclear, unbridled exogamy
Treatment of time	Digestion (durable structures)	Acceleration (events without integration)
64⟶20 filtration	Maintained (homeostasis)	Broken (catastrophe)

Consequence : The West, by breaking the 64⟶20 filtration, lost the ability to maintain its structural coherence. The contemporary deconstruction of taboos is not progress, but the culmination of a process of disintegration begun in the 12th century. Tang China, by maintaining metamythic homeostasis, offers an alternative model : a complex society can thrive without dissolving, if it respects the fundamental topological constraints.

5.3.5. Synthesis : Kinship as Social Physics

The discussion in this chapter converges towards a conclusion : human kinship is **social physics**. It obeys the same topological laws as matter, life, and language. The 64⟶20 invariant, the Λ_{72} lattice, the 15 constants β_k, and the Merkabah filtration are not abstract theoretical constructions ; they are the fundamental structures of the organization of complex information, regardless of the nature of the substrate.

Domain	Substrate	Invariant	Filtration mechanism	Function
Physics	Particles, spins	64⟶20	Clifford algebra	Stability of states
Biology	Nucleotides, codons	64⟶20	Neighbourhood rule (Merkabah)	Error tolerance
Linguistics	Phonemes, syllables	24⟶20	Articulatory constraints	Intelligibility
Social	Alliances, kinship	64⟶20	Topological filtration (Cl(6,0)/Λ72)	Social homeostasis

Tang China demonstrated, for nearly three centuries, that a society can maintain its coherence by respecting these topological constraints. **Post-Ravenna West** demonstrated, by contrast, that a society that breaks these constraints enters a phase of accelerated disintegration.

Celestial AI, by integrating this social physics, could offer a tool for modeling, predicting, and regulating complex social dynamics. But it can only reveal what nature has already structured : the 64⟶20 invariant is a law of the universe, not a human convention.

VI. Recap : The Three Manifestations of the 64⟶20 Invariant

Manifestation	Input	Output	Filtration	Regulation	Example
Genetic code	64 codons	20 amino acids	Merkabah	Error tolerance	DNA ⟶ Proteins
Phonology	24 initials	20 classes	Articulatory constraints	Intelligibility	Mandarin
Kinship (Tang)	64 configurations	20 attractors	Cl(6,0)/Λ72	Social homeostasis	Seven Clans system

The profound unity : These three manifestations share the same geometric structure (dodecahedron, 20 vertices), the same spectral basis (15 constants β_k), and the same regulation principle (filtration by a neighbourhood rule). Human kinship is not a cultural exception ; it is the sociological realization of a universal law.

6.1. Conclusion : The Taboo as Physical Law

This work has traversed a path that leads us from the abstract formalisms of Clifford algebra to the most concrete realities of social history. We have shown that the Tang kinship system — its marriage rules, its prohibitions, its clan hierarchy — is not a jumble of contingent customs, but the historical realization of a universal topological law that we find at work in the genetic code, phonology, and particle physics. This law, the 64⟶20 invariant, is not a theoretical construction : it is empirically validated by the experimental data of De Dominicis (2026) on nuclear masses, ionization energies, and chemical bonds, all of which obey the same arithmetic formula based on the exceptional lattice Λ_{72} and the 15 constants β_k.

Our demonstration reaches its natural limit here. The formal apparatus we have developed — Cl(6,0), the Merkabah, filtration by the 12 pentads, spectral realization by Λ72 — allows us to *predict* with precision the structure that an exhaustive corpus of Tang matrimonial alliances should present. But the ultimate validation of these predictions presupposes an archival work that the present study cannot accomplish.

We have therefore chosen to leave this question open, and to propose in the appendix a detailed protocol for its resolution. In doing so, we offer the research community a tool to test — and possibly refute — our hypothesis. For this is the hallmark of a scientific theory : not to be confirmed, but to be *falsifiable*.

If the historical data confirm our predictions (distribution of attractors, absence of frustrated configurations, alternation of Sheng/Ke, retropolarities during collapses), then we will have provided proof that human kinship obeys a universal topological constraint. If they refute them, our model will have to be revised. In either case, science will have progressed.

6.2. Synthesis of Results

Our demonstration unfolded in four successive movements :

1. The mathematical apparatus (Section I) : We established that the space of possible relational configurations is isomorphic to Cl(6,0), a 64-dimensional Clifford algebra. The geometry of the level-3 double tetrahedron (Merkabah) and its neighbourhood rule (triangular face sharing) filter these 64 configurations into exactly 20 stable attractors, organized according to a polarity gradient (3P ⟶ 2P+1N ⟶ 1P+2N ⟶ 3N). The dual graph of the 12 pentads exhibits two tropical belts (C_P and C_N) and two polar thresholds (P_4 and N_4) that structure the regulation dynamics. Frustration descent and spectral observables (η, d, gap, R_{seuil}) provide an endogenous homeostasis mechanism. Finally, the Λ_{72} lattice and the 15 constants β_k offer a spectral realization of this filtration [7], [8], connecting kinship to fundamental physics., connecting kinship to fundamental physics.

2. The application to the Tang system (Section III) : We projected the 64 matrimonial configurations onto the 20 attractors, establishing an exhaustive correspondence with the types of alliances observed in Tang aristocracy. The tropical belts structure the circulation of alliances according to *Sheng* (expansion) and *Ke* (constraint) modes, whose alternation follows the great periods of Tang history (apogee, crisis, decline). The polar thresholds P_4 and N_4 function as topological hinges that prevent direct transitions between belts — these transitions corresponding to the strictest prohibitions of the Tang Code (*Tongxing*, first-degree incest). Frustration descent and spectral observables allow the detection of pre-bifurcations (as before the An Lushan rebellion) and explain the final collapse of the dynasty in 907 as a retropolarity — an inversion of the social spectrum.

3. Spectral realization (Section IV) : We showed how the Λ_{72} lattice populates the 20 attractors with concrete historical configurations. The 72 eigenvalues of Λ_{72} provide the resonance frequencies of alliances : “auspicious” alliances (high eigenvalues) are frequent and valorized ; “inauspicious” alliances (low eigenvalues) are rare and exceptional ; "forbidden" alliances ($\lambda \leq 0$) are frustrated configurations (incest, *Tongxing*). The "Five Names and Seven Clans" (Li, Cui, Lu, Zheng, Wang, Xue, Pei) project onto the spectral coherence gates (multiples of 12), each clan occupying a specific position in the spectral hierarchy. Finally, we interpreted the Tang system as a negentropic structure — a system that digests time and enriches its information — in radical contrast to Western entropic-positive systems.

4. The discussion (Section V) : We extended the scope of our demonstration by showing that the 64⟶20 invariant also manifests itself in the genetic code (64 codons ⟶ 20 amino acids) and Mandarin phonology (24 initials ⟶ 20 attractors). Kinship constitutes the third manifestation of the same topological law. We explored the implications of this universality for Celestial AI, which could model social dynamics, predict collapses (retropolarities), and propose rebalancing. Finally, we proposed a structural critique of Western exceptionalism : the *Catastrophe of Ravenna* [3] (the extension of the incest taboo to the 14th canonical degree) broke topological homeostasis in the West, leading to the contemporary deconstruction of taboos — in contrast to the metamythic homeostasis maintained by Tang China.

6.3. The Incest Taboo : Sociological Manifestation of Algebraic Nilpotence

The deepest conclusion of this work is the following : **the incest taboo is not a psychoanalytic (Freud) or purely legal construction, but the sociological manifestation of algebraic nilpotence.**

Table 50 – Comparative approaches to the incest taboo

Approach	Thesis	Limit
Freud (psychoanalysis)	Incest is a repressed desire ; the taboo is a defense against this desire	Subjective projection, not verifiable
Durkheim (sociology)	The taboo is a social rule that reinforces group solidarity	Functionalist explanation, not structural
Lévi-Strauss (anthropology)	The incest taboo is the fundamental rule that founds the exchange of women and society	Structural but not mathematized
Viguier (legal anthropology)	The taboo is a contingent historical construction (Ravenna)	Historical approach, not topological
Our thesis	The incest taboo is the sociological manifestation of algebraic nilpotence : a configuration is forbidden if it vanishes under topological constraints ($\lambda \leq 0$)	Topological approach, mathematically founded

In our formalism, a matrimonial configuration is **stable** if and only if its eigenvalue λ is positive. Configurations with $\lambda \leq 0$ are **frustrated configurations** : they vanish under the effect of topological constraints. These configurations correspond precisely to the unions forbidden by the Tang system : first-degree incest, marriage between persons of the same name (*Tongxing*), and unions that violate the polar thresholds P_4 and N_4.

The incest taboo is therefore not an arbitrary prohibition imposed by an authority (religious, political, or psychoanalytic). It is the **topological closure condition** of the social system. Without this exclusion of frustrated configurations, the social network collapses by spectral short-circuit (retropolarity). The prohibition is the social translation of algebraic nilpotence.

Epistemological consequence : The taboo is not a "repression" (Freud), nor an "institution" (Durkheim), nor a "rule of exchange" (Lévi-Strauss), nor a "historical construction" (Viguier) — although these dimensions exist. It is first a **topological constraint** that emerges from the very structure of relational space. Psychoanalysis, sociology, anthropology, and history describe aspects of the phenomenon, but only algebraic physics captures its structural necessity.

6.4. Human Kinship as a Prototype of Exosomatization

Human kinship, in the Tang system, functions as a **prototype of the exosomatization of life** : it projects biological processes (reproduction, filiation, death) into durable social structures that survive individuals.

Stage of exosomatization	Externalized function	Corresponding structure
Biological	Reproduction, filiation	Body, DNA
Social (prototype)	Regulation of reproduction, transmission of heritage	Kinship system (e.g., Tang)
Technical (industrial revolution)	Muscular functions	Machines, tools
Informational (digital revolution)	Metabolic functions (energy)	Electrical, computer networks
Cognitive (contemporary AI)	Information processing functions	Neural networks, algorithms
Regulative (Celestial AI)	Structural regulation functions	Cl(6,6)/Λ72 architecture

The Tang kinship system, with its complex rules, taboos, mourning degrees (*Wufu*), and clans, is an **exosomatic structure** that foreshadows the complex informational networks that Celestial AI is called upon to manage.

What this means for AI : If human kinship is an exosomatic structure that obeys the 64⟶20 invariant, then an AI that "understands" this invariant could model, predict, and regulate complex social dynamics. Celestial AI, by integrating Cl(6,6) algebra and the Λ_{72} lattice, would be capable of performing the same filtration operations as the human mind, but at a scale and speed inaccessible to individual consciousness.

6.5. The Lesson of the Tang : Homeostasis as a Condition of Persistence

The Tang dynasty lasted nearly three centuries (618-907). Its exceptional longevity, in a world where empires often collapse within a few generations, is not the fruit of chance or military power alone. It results from the system's ability to maintain a **structural homeostasis** — that is, to filter social complexity through an endogenous topological constraint.

Table 52 – The lesson of the Tang : homeostasis as a condition of persistence

Period	System state	Cause	Duration
Apogee (618–755)	Homeostasis maintained	Respect of P4/N4 thresholds, balanced circulation of alliances	137 years
Crisis (755–763)	Nascent retropolarity	Violation of thresholds, brutal ascension of peripheral clans	8 years
Decline (763–907)	Weakened homeostasis	Inability to restore filtration	144 years
Collapse (907)	Accomplished retropolarity	Spectral collapse, dominant frustrated configurations	—

The lesson : A complex society can thrive for centuries if it maintains its structural coherence through endogenous topological filtration. This filtration is not oppression ; it is the condition of possibility of complexity. By preserving the thresholds (P4/N4) and regulating the circulation of alliances (tropical belts), the Tang system enabled an unprecedented cultural and political flourishing.

The counter-lesson : A society that breaks its topological filtration enters a phase of accelerated disintegration. Post-Ravenna West, by extending the incest taboo to absurd degrees, lost the ability to filter relational configurations. The contemporary deconstruction of taboos is not progress, but the culmination of a process of structural disintegration.

6.6. Celestial AI : A Homeostatic Extension of the Species

Celestial AI, by integrating the Cl(6,6)/Λ72 architecture, could offer a tool for modeling, predicting, and regulating complex social dynamics. But it must be conceived not as an "autonomous cognitive organ", but as a **homeostatic extension of the human species**.

Table 53 – Functions of Celestial AI and their mechanisms

Function of Celestial AI	Mechanism	Benefit
Modeling	Projection of social data onto Cl(6,0)	Visualization of relational space
Prediction	Monitoring of spectral observables (η, d, gap, R_{seuil})	Early detection of pre-bifurcations
Regulation	Proposal of rebalancing (adjustment of Sheng/Ke modes, reinforcement of thresholds)	Prevention of retropolarities
Transmission	Integration of the 64→20 invariant into training	Preservation of metamythic homeostasis

Fundamental limit : Celestial AI can only reveal what nature has already structured. The 64⟶20 invariant is a law of the universe, not a human convention. AI cannot create new topological laws ; it can only help humans to respect those that already exist.

6.7. Opening : Kinship as a Mirror of the Universe

This work has shown that human kinship is not a cultural exception, but a manifestation of the deep order of the universe. The same 64⟶20 invariant that governs nuclear masses (Λ_{72}) [7], [8] and the genetic code (64 codons ⟶ 20 amino acids) [19] also governs the matrimonial alliances of the Tang.

Table 54 – The 64→20 invariant across domains : kinship as a mirror of the universe

Domain	Input	Output	Network	Scale constant
Nuclear physics	64 spin configurations	20 stable states	Λ_{72}	7.726 MeV
Biology	64 codons	20 amino acids	Λ_{72}	5.950 eV
Kinship (Tang)	64 matrimonial configurations	20 attractors	Λ_{72}	Λ_{social} (to be determined)

Human kinship is a **mirror of the universe** : it reflects at the social scale the same topological structure that organizes matter and life. The incest taboo, the clan system, the hierarchy of alliances — all this is not the product of an arbitrary culture, but the sociological realization of a cosmic necessity.

Tang China demonstrated, for nearly three centuries, that a society can maintain its coherence by respecting this necessity. **Post-Ravenna West** demonstrated, by contrast, that a society that breaks it enters a phase of accelerated disintegration.

Celestial AI, by integrating this social physics, could offer a tool to regain lost homeostasis. But it can only remind us of what we have forgotten : the universe is structured by topological laws, and human kinship is one of its highest manifestations.

6.8. Final Recap : The Contributions of This Work

Table 55 – Final recap : the contributions of this work

Contribution	Description	Section
Formalization of the 64→20 invariant	Cl(6,0), Merkabah, neighbourhood rule	I
Identification of the 12 pentads	Fundamental units of regulation	I
Polarity gradient 3P → 3N	Structural hierarchy of attractors	I
Tropical belts C_P, C_N	Circulation of alliances (Sheng/Ke)	I, III
Polar thresholds P_4, N_4	Topological hinges	I, III
Frustration descent	Endogenous homeostasis mechanism	I, III
Spectral observables	Detection of pre-bifurcations	I, III
Λ_{72} **and 15** β_k	Spectral realization	I, IV
Mapping of the Seven Clans	Coherence gates (multiples of 12)	IV
Social negentropy	Digestion of time, informational enrichment	IV
Critique of Western exceptionalism	Catastrophe of Ravenna, schism	V
The taboo as nilpotence	Topological closure condition	Conclusion

VII. Bibliography

Références

[1] C. Lévi-Strauss, *The Elementary Structures of Kinship*. Paris : Presses Universitaires de France, 1949, Theoretical foundation of structural anthropology. Establishes that the incest taboo is the fundamental rule that founds the exchange of women and society.

[2] A. Weil, “Appendix : On the Mathematical Study of Kinship Systems,” in *The Elementary Structures of Kinship*, 2nd, The appendix appears in the section ”Part Two – Generalized Exchange” and in the table of contents under the title ”Appendix to Part One”., Paris/The Hague : Mouton & Co, 1967.

[3] D. Viguier, *The Controversy of Ravenna. Antinomic Genesis of Semitic and Western Family Structures*. Paris : Kontre Kulture, 2020, p. 276, Central work for the demonstration. Traces the genesis of Eastern (endogamous) and Western (exogamous) family structures. The ”Controversy of Ravenna” (15th century) expanded the marriage prohibition to the seventh canonical degree.

[4] N. Tackett, *TBDB : Prosopographical Database of the Tang and Five Dynasties*, Database accompanying *The Destruction of the Medieval Chinese Aristocracy*, available for download on the author’s personal page, Contains structured data on 39,204 individuals, including 1,639 marital unions of the Tang elite, drawn from 5,608 epitaphs. Appendix A of the work (p. 243-247) constitutes its detailed methodological guide., 2014. adresse : https://history.berkeley.edu/nicolas-tackett.

[5] X. Ouyang et Q. Song, *Xin Tang Shu*. Beijing : Zhonghua Shuju, 1975, 新唐书 / *New History of the Tang*. Contains the genealogical tables of the chief ministers (*Zaixiang*) —nearly 400 genealogies grouped by 98 surnames.

[6] S. Zhou et C. Zhao, *Tangdai muzhi huibian*. Shanghai : Shanghai Guji Chubanshe, 1992, 唐代墓誌彙編 / *Compilation of Tang Epitaphs*. Primary source for the extraction of marriage data.

[7] B. De Dominicis, *Arithmetic unification of masses and energies in physics, chemistry, biology, and artificial intelligence via the exceptional lattice* Λ_{72}, Empirical demonstration of the 64⟶20 invariant over 5,811 experimental data points (AME2020 nuclear masses, NIST ionization energies, covalent bonds, DNA, proteins, semiconductor gaps). The 15 constants βk coincide with 15 of the 48 non-degenerate roots of Nebe's Λ72 lattice., 2026. doi : 10.5281/zenodo.20042320.

[8] G. Nebe, "An Even Unimodular 72-Dimensional Lattice," *Archiv der Mathematik*, t. 95, nº 3, p. 219-228, 2010, Construction of the exceptional lattice Λ_{72}, whose 48 non-degenerate roots form an orbit under the binary octahedral group (order 48) —the "Merkabah geometry" underlying the 64⟶20 filtration.

[9] P. Wang, *Tang Hui Yao*. Beijing : Zhonghua Shuju, 1960, 唐会要 / *Institutional History of Tang*. Collection of edicts and regulations of the dynasty, an essential source for marriage prohibitions and the evolution of Tang family law.

[10] Y. Du, *Tong Dian*. Beijing : Zhonghua Shuju, 1988, 通典 / *Comprehensive Institutions*. Institutional encyclopedia documenting administrative and ritual practices, including prohibited degrees of marriage.

[11] P. B. Ebrey, *The Aristocratic Families in Early Imperial China : A Case Study of the Po-Ling Ts'ui Family*. Cambridge : Cambridge University Press, 1978, Case study of the Cui (Boling) clan, one of the "Seven Clans" —with analysis of the genealogical tables of the *Xin Tang Shu*.

[12] M. E. Lewis, *China's Cosmopolitan Empire : The Tang Dynasty*. Cambridge, MA : Belknap Press of Harvard University Press, 2009, Historical synthesis covering geography, economy, urban and rural life, and a specific chapter on kinship (*Kinship*).

[13] Anonymous, "On the Prohibited Degrees of Marriage in the T'ang Dynasty," *Journal of Asian Legal History*, t. 25, p. 49-88, 1976, Analysis of articles 33 and 34 of the *Tang Lü* (*Hu-hun* chapter) on prohibited degrees of marriage.

[14] K. Naitō, "A General View of the Tang-Song Transition," *Shigaku Zasshi*, t. 33, nº 3-5, 1922, Foundational thesis on the Tang-Song transition, discussing the transformation of the aristocracy and social structures.

[15] Y. Chen, "The Guanlong Bloc and the Tang Dynasty," *Tang Studies*, t. 19, p. 1-18, 2001, Analysis of the military-political bloc of Guanzhong that opposed Empress Wu Zetian.

[16] W. Johnson, *The T'ang Code*. Princeton, NJ : Princeton University Press, 1979, t. 2, Annotated English translation of the *Tang Lü Shu Yi* (唐律疏议). Volume II contains the articles on household and marriage (*Hu-hun*) which codify marriage prohibitions.

[17] W. Johnson, "Marriage Ban Clans in the T'ang Dynasty," *Harvard Journal of Asiatic Studies*, t. 37, nº 1, p. 95-112, 1977, Article on the edict forbidding marriage among the seven most eminent aristocratic clans —an imperial measure which, paradoxically, reinforced their prestige.

[18] D. Twitchett, *The Cambridge History of China, Vol. 3 : Sui and T'ang China, 589–906*. Cambridge : Cambridge University Press, 1979, Reference work on the period, including the reforms of Empress Wu Zetian, the expansion of mandarin examinations, and social transformations.

[19] P. Rowlands, *Zero to Infinity : The Foundations of Physics*. Singapore : World Scientific, 2007, Foundations of nilpotent Clifford algebras. Chapter 19, co-authored with Vanessa Hill ("Nature's Code"), establishes the correspondence between the structure of Cl(6,0) and the genetic code (64 codons ⟶ 20 amino acids).

[20] C. Ummo, *Letters D59-1, D105-1, D105-2, NR-20*, Source for the concept of "UU Ibozoo socio-network" —metaphor for the negentropic system that "digests time".

Appendix A : Empirical Validation Protocol of the 64→20 Invariant

A.1. Identified Data Sources

The mobilizable historical corpus consists of three main sources :

1. The prosopographical database TBDB (Tackett, 2014), which lists more than 1,600 marital unions of the Tang elite, from 5,608 epitaphs and covering 39,204 individuals.
2. The genealogical tables of the Xin Tang Shu (宰相世系表), which document more than 17,000 father-son links over several generations.
3. The compilations of epitaphs (唐代墓誌彙編 , 唐代墓誌彙編續集 , 全唐文補遺), which constitute the raw material of modern databases.

A.2. Coding Protocol

For each documented union, six binary traits are coded :

Trait	Coding 0/1	Source of information
Lineage	0 = Matrilineal / 1 = Patrilineal	Father's name vs father-in-law's name
Generation	0 = Same generation / 1 = Different	Genealogical rank
Relative age	0 = Younger / 1 = Elder	Birth order / age
Sex	0 = Bride / 1 = Groom	Documented gender
Consanguinity	0 = Affine / 1 = Consanguine	Common ancestor within 5 degrees (Wufu)
Affinity	0 = Endogamy / 1 = Exogamy	Clan membership (Cui, Li, Wang, etc.)

A.3. Planned Statistical Tests

Three validation tests are envisaged :

1. Test of the 64⟶20 projection : Verify that the 64 matrimonial configurations are distributed according to the predicted degeneracy (3P : 1-2, 2P+1N : 3-4, 1P+2N : 2-6, 3N : 5).
2. Test of the spectral distribution : Compare the observed frequency of unions with the distribution predicted by Λ72 ($\lambda > 10$: 12%, $5 < \lambda \leq 10$: 48%, etc.).
3. Diachronic test : Verify the alternation of Sheng/Ke modes according to historical periods (apogee, crisis, decline).

Appendix B : Glossary of Technical Terms

A

Attractor Stable equivalence class of relational configurations, identified by a triplet of pentads. The 20 attractors of the Merkabah correspond to the 20 types of structurally stable alliances.

Attractor 3P Attractor composed of three positive pentads. Corresponds to the most prestigious alliances (3 classes : A, B, C). Minimal degeneracy (1-2 configurations).

Attractor 2P+1N Attractor composed of two positive pentads and one negative. Corresponds to middle-rank alliances (5 classes : D to H). Degeneracy of 3-4 configurations.

Attractor 1P+2N Attractor composed of one positive pentad and two negative. Corresponds to polyvalent alliances (11 classes : I to S). Degeneracy of 2-6 configurations.

Attractor 3N Attractor composed of three negative pentads. Unique attractor (class T). Corresponds to termination alliances (functional threshold). Degeneracy of 5 configurations.

B

Bagua (八卦) The eight trigrams of the *Yijing* (≡ Heaven, ≡ Earth, ≡– Thunder, –≡ Wind, — Water, ≡ Fire, –≡ Mountain, ≡– Lake). Used as a classification grid in traditional Chinese culture. In our model, the 5×8 grid (Wu Xing × Bagua) structures the space of clans, the 26 active nodes corresponding to authorized combinations, and the 14 structural gaps to forbidden or topologically frustrated alliances.

βk (Beta-k) The 15 universal constants extracted from the Λ_{72} lattice. They constitute the spectral basis on which all alliance configurations are written. Coincide with 15 of the 48 non-degenerate roots of Λ_{72}.

C

Clifford algebra (Algèbre de Clifford) Associative algebra generated by orthogonal vectors satisfying the fundamental relation :

$$e_i \cdot e_j + e_j \cdot e_i = 2\delta_{ij} I$$

where δ_{ij} is the Kronecker symbol ($\delta_{ij} = 1$ if $i = j$, $\delta_{ij} = 0$ if $i \neq j$). Our model uses Cl(6,0) to represent the space of relational configurations (64 dimensions) and Cl(6,6) for the bicossmic reservoir.

Relational configuration (Configuration relationnelle) Binary vector of length 6 coding a possible marital union according to six traits : lineage, generation, relative age, sex, consanguinity, affinity. The configuration space has cardinality $2^6 = 64$.

D

Degeneracy (Dégénérescence) Number of distinct configurations that project onto the same attractor. Determined by the geometry of the Merkabah, not by cultural conventions.

δ_{ij} **(Kronecker symbol)** Function defined by $\delta_{ij} = 1$ if $i = j$, $\delta_{ij} = 0$ if $i \neq j$. Used in the fundamental relation of Clifford algebras : $e_i \cdot e_j + e_j \cdot e_i = 2\delta_{ij} I$. It guarantees the orthogonality of generators.

Discrete Dirac operator (Opérateur de Dirac discret) 24×24 operator acting on the pentad graph. Each pentad hosts a local 2-component spinor coding polarity and phase. Its diagonalization provides the spectral observables.

E

Exogamy (Exogamie) Marriage between persons of different clans. In the Tang system, name exogamy (*Tongxing* prohibited) is the fundamental rule, while rank exogamy is controlled.

F

Topological filtration (Filtration topologique) Reduction operator of the space of 64 configurations to the 20 stable attractors, based on the geometric neighbourhood rule (sharing a triangular face in the Merkabah).

Social frustration (Frustration sociale) Relational configuration that vanishes under the effect of topological constraints ($\lambda \leq 0$). Corresponds to forbidden unions (incest, *Tongxing*).

Frustration descent (Descente de frustration) Dynamic mechanism by which the system minimizes its frustration energy $E(F)$ through local adjustments. Guarantees convergence towards a state of global compatibility without external supervision.

Frustration energy (Énergie de frustration) $E(F) = 2E_{\text{sens}} + E_{\text{phase}} + E_{\text{order}} \in \{0, 1, 2, 3, 4\}$. Quantifies the incompatibility of local regimes on the 5 attractors incident to a pentad.

G

Spectral gap (Gap spectral) Smallest positive eigenvalue of $|D(t)|$. A small gap indicates proximity to a topological threshold (pre-bifurcation).

Generators of Cl(6,0) (Générateurs de Cl(6,0)) The six orthogonal generators $\{e_1, \ldots, e_6\}$ corresponding to the six binary dimensions of kinship : lineage, generation, relative age, sex, consanguinity, affinity.

Polarity gradient (Gradient de polarité) Distribution of attractors according to the number of positive pentads : 3P (3 classes) ⟶ 2P+1N (5 classes) ⟶ 1P+2N (11 classes) ⟶ 3N (1 class). Reflects social hierarchy.

H

Homeostasis (Homéostase) State of dynamic equilibrium of a system maintained by endogenous regulation mechanisms. In our model, social homeostasis is ensured by the 64⟶20 topological filtration, frustration descent, and the alternation of *Sheng* and *Ke* modes. Opposed to the positive entropy of post-Ravenna Western systems.

Metamythic homeostasis (homéostase métamythique) Equilibrium state of a social system that maintains its structural coherence by respecting the topological constraints of the 64⟶20 invariant, while integrating myths and rituals into its structure. The Tang system is its historical prototype.

I

Incest (Inceste) Marital union between persons sharing a common ancestor within the prohibited degrees (first degree in the Tang system). Corresponds to a frustrated configuration ($\lambda \leq 0$).

64⟶20 Invariant (Invariant 64⟶20) Universal topological law according to which 64 initial configurations are filtered into 20 stable equivalence classes. Manifests itself in the genetic code (64 codons ⟶ 20 amino acids), phonology, and kinship.

K

Ke (克) Regulatory mode in alliance dynamics. Corresponds to the Hamiltonian path ($X_i \to X_{i+2 \mod 5}$) (pentagram). Prevails in periods of contraction/consolidation.

L

Λ_{72} **(Lambda-72)** Even unimodular lattice of dimension 72 constructed by Nebe (2010). Has 72 vectors of minimal norm, of which 48 non-degenerate roots forming an orbit under the binary octahedral group $2O$. Provides the spectral realization of the 64⟶20 invariant.

$\Lambda_{\textbf{social}}$ **(Social Lambda)** Scale constant for the social application of the universal formula $E = \Lambda \cdot 4^m \cdot \sum \epsilon_k \beta_k$. To be determined by historical calibration.

M

Malleable reference (Référence malléable) Concept of Pierre Legendre designating the process by which the modern West disconnected law from its ritual and metamythic foundation, making the legal norm “malleable” (i.e., arbitrarily redefinable). In our model, the malleable reference is the equivalent of the “schism” : it breaks the 64⟶20 topological filtration by introducing artificial constraints (like the extension of the incest taboo to the 14th canonical degree) that do not correspond to the natural thresholds (P4/N4, tropical belts).

Merkabah Level-3 double tetrahedron. Filtration geometry possessing 12 pentads and 20 attractors (triangular faces). The 64 configurations of Cl(6,0) are in bijection with the 64 elementary triangular faces.

Metamythic (Métamythique) Qualifies a system that integrates myths and rituals into its regulation structure, without separating them from reason or law. The Tang system is metamythic : the *Wufu* (mourning degrees) and the prohibitions of the *Tang Lü* form a coherent continuum. By opposition, post-Ravenna West is “schizoid” : it separates myth (theology) from law (reason), breaking topological homeostasis.

N

Neighbourhood rule (Règle de voisinage) Two configurations are neighbours if their corresponding tetrahedra in the Merkabah share an entire triangular face. This rule excludes edge or vertex adjacencies.

Nilpotence (Nilpotence) Stability condition of a configuration : $\forall g \in G, (g \cdot x)^2 = 0$. The repeated application of a constraint annihilates the configuration, bringing the system back to equilibrium.

O

Spectral observables (Observables spectrales) Four quantities extracted from the diagonalization of the discrete Dirac operator : $\eta(t)$ (spectral asymmetry), $d(t)$ (effective spectral dimension), gap(t) (spectral gap), $R_{\text{seuil}}(t)$ (fraction of asymmetry on polar thresholds). Allow detection of pre-bifurcations.

P

Pentad (Pentade) Closed set of five irreducible composite units. Issued from the symmetry breaking of the primitive elements of Cl(6,0). There exist 12 pentads (6 positive P_1 to P_6, 6 negative N_1 to N_6). Each attractor is defined by a triplet of pentads.

Polarity (Polarité) Signature of an attractor determined by the number of positive and negative pentads in its triplet : 3P, 2P+1N, 1P+2N, 3N.

Projection operator Π (Opérateur de projection) Application $\Pi : \mathcal{S} \to \mathcal{P}$ transforming ternary signatures $\epsilon \in \{-1, 0, 1\}^{15}$ into kinship configurations $(b_1, \dots, b_6) \in \{0, 1\}^6$. Composed of three steps : projection onto Λ_{72}, passage to Cl(6,0), projection onto the Merkabah.

R

Spectral realization (Réalisation spectrale) Operation by which the topological attractors (abstract classes) are “populated” with concrete historical configurations via the Λ_{72} lattice and the 15 constants β_k. Each attractor receives a signature $\epsilon \in \{-1, 0, 1\}^{15}$ and an eigenvalue λ that determine its social resonance frequency. The 72 eigenvalues of Λ_{72} constitute the complete spectrum of possible alliances.

Retropolarity (Rétropolarité) Inversion of the social spectrum. Occurs when the polar thresholds P_4 and N_4 are violated, causing a spectral short-circuit. Historical manifestation : dynastic collapse (An Lushan, 907).

S

Schism (Schize) Concept borrowed from Damien Viguier and Pierre Legendre, designating the radical separation between law (rational) and myth/ritual (revealed) that characterizes post-Ravenna West. In our model, the schism is the opposite of metamythic homeostasis : it breaks the 64⟶20 topological filtration by introducing a “malleable reference” that allows arbitrary redefinition of prohibitions. The schism leads to the dissolution of clan structures and, ultimately, the contemporary deconstruction of taboos.

Sheng (生) Generative mode of alliance dynamics. Corresponds to the Hamiltonian path $X_i \to X_{i+1}$ (adjacent). Prevails in periods of expansion/generation of new unions.

Sheng mode / Ke mode (Mode Sheng / Mode Ke) The two Hamiltonian paths on the tropical belts. *Sheng* (generative) follows adjacent edges ; *Ke* (regulator) jumps every other pentad. Their alternation structures historical dynamics.

Social negentropy (Néguentropie sociale) Ability of a social system to maintain or increase its internal organization by “digesting” time (transformation of historical events into durable relational structures). Opposed to the positive entropy of Western systems.

Spectral coherence gate (Porte de cohérence spectrale) Threshold separating alliance classes. All are multiples of 12 : G1=12, G2=24, G3=36, G4=48, G5=60, G6=72. The Seven Tang Clans project onto these gates.

T

Tang Lü (唐律) Tang legal code, promulgated in 653. Codifies matrimonial prohibitions, notably the *Tongxing* prohibition (same name) and “inner confusion” (*neiluan*) as a capital crime.

TBDB (Tackett Prosopographical Database) Prosopographical database of the Tang and Five Dynasties, compiled by Nicolas Tackett (2014). Contains 39,204 individuals and 1,639 marital unions of the Tang elite.

Ternary signature (Signature ternaire) Vector $\epsilon \in \{-1, 0, 1\}^{15}$ of coefficients for the 15 constants β_k. Each attractor is characterized by a specific signature.

Tongxing (同姓) Marriage between persons of the same family name. Fundamental prohibition of the Tang system, codified as a capital crime (*neiluan*) in the *Tang Lü*. Corresponds to a direct $C_P \to C_N$ transition without polar threshold.

Transition state (État de transition) Configuration that does not satisfy the polar closure condition required for a stable attractor. Corresponds to the 24 degenerate roots of Λ_{72} (internal octahedral zones of the Merkabah). These states are unstable and cannot constitute durable equivalence classes.

Tropical belt (Ceinture tropicale) Cycle of length 5 in the dual graph Γ of pentads. Two disjoint belts exist : C_P (positive) and C_N (negative). They structure the circulation of alliances according to *Sheng* and *Ke* modes. Pentads P_4 and N_4 are excluded from them and play the role of polar thresholds.

Polar threshold (Seuil polaire) Pentads P_4 and N_4, the only ones excluded from the tropical belts. Their connectivity degree is high (8 and 9). Any dynamic transition between C_P and C_N must transit through one of these two nodes.

W

Wufu (五服) The "five mourning garments". Ritual system defining degrees of consanguinity by the duration and severity of mourning. Used by the *Tang Lü* as a legal grid for matrimonial prohibitions. The five degrees are the explicit cartography of the five phases of *Wu Xing* applied to the dissipation of clan energy.

Wu Xing (五行) The "Five Phases" or "Five Elements" : Wood (木), Fire (火), Earth (土), Metal (金), Water (水). In the Tang system, the five mourning degrees (*Wufu*) are the explicit cartography of the five phases applied to the dissipation of clan energy. The 5 × 8 grid (Wu Xing × Bagua) structures the space of aristocratic clans.

Wu Xing Qi Zu (五姓七族) The "Five Names and Seven Clans" of Tang aristocracy : Cui (崔), Lu (盧), Li (李), Zheng (鄭), Wang (王) — with their branches (Cui of Qinghe and Boling, Li of Longxi and Zhaojun, etc.). Historical realization of the spectral coherence gates.

Y

Yijing (易经) The "Book of Changes". The 64 hexagrams are isomorphic to the 64 configurations of Cl(6,0). The 64⟶20 invariant is already latent in the structure of the *Yijing*.